

Homochiral ideas

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Homochiral python code on github

- <https://github.com/marimperorclerc/homochiral>

python code for drawing clusters with $\text{Tile}(a,b)$ in svg graphics

References

Code is based on the drawsvg package:

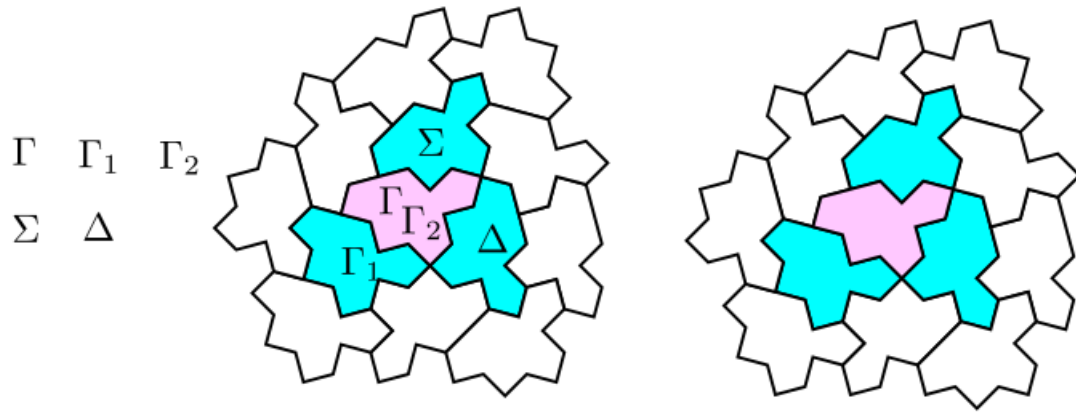
<https://pypi.org/project/drawsvg/>

And the python code with alternate chirality for inflation for spectre tiling:

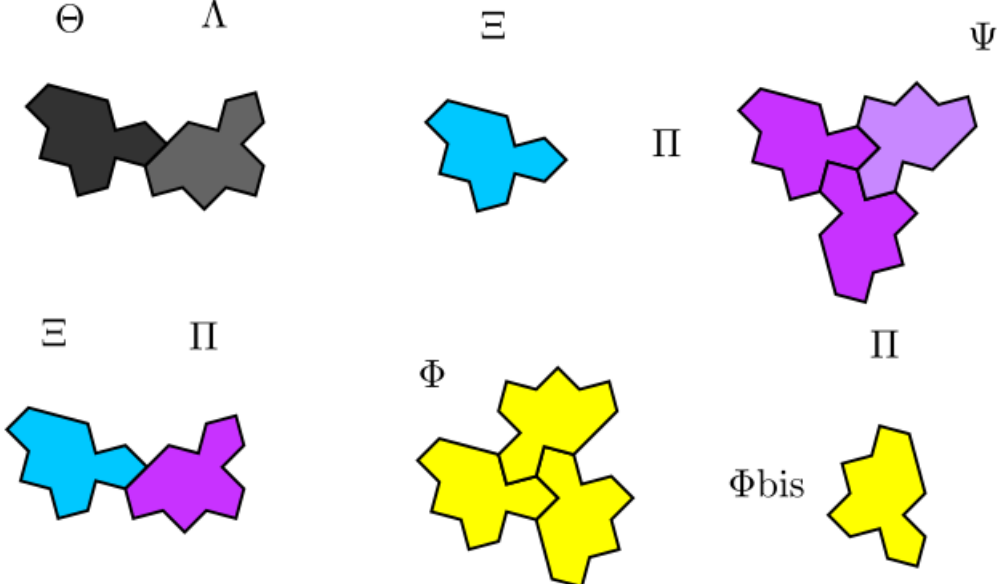
<https://github.com/shrx/spectre/blob/master/spectre.py>

- Adapted here (spectre-colored-tiles.ipynb) with different color choices
- See some .svg files in \svg-spectre folder

Identification of the greek letters for tiles



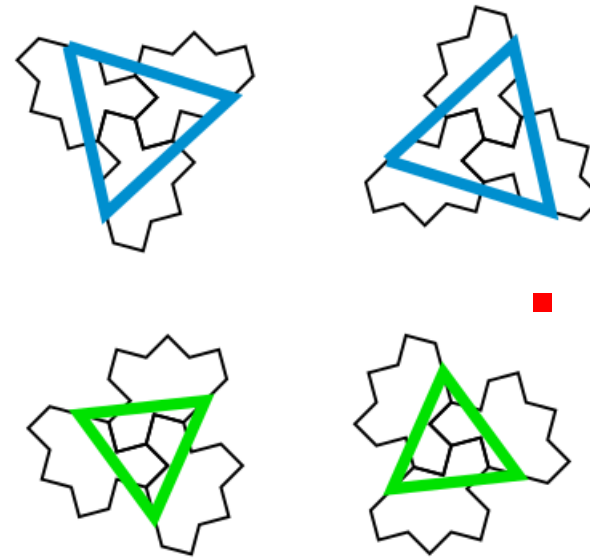
- $\backslash\theta$ and $\backslash\lambda$ always in couple



- All $\backslash\chi$ tiles are linked to a $\backslash\pi$ tile and consequently to a large star.

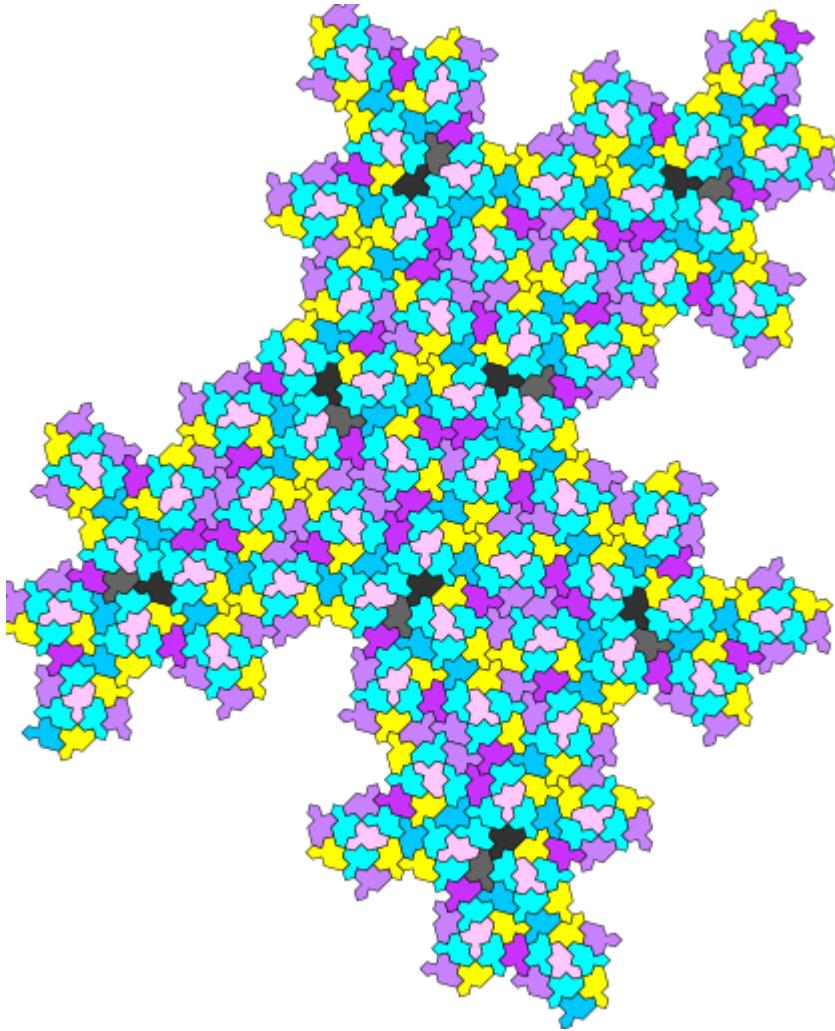
- Consider a cluster of four tiles instead of 3 for horn in the anatomy ? (by adding the $\backslash\Delta$ tile)

- Large stars correspond to all $\backslash\pi$ and $\backslash\psi$ tiles



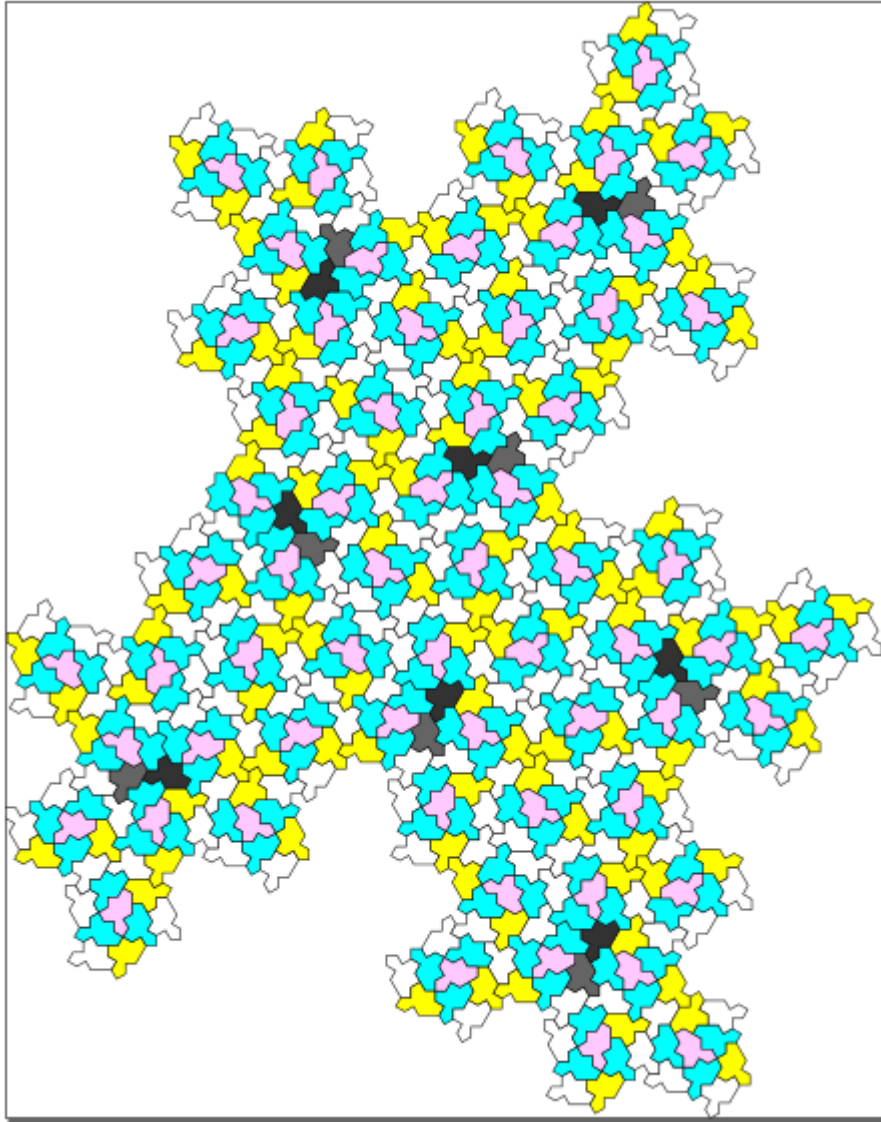
- $\backslash\phi$ tiles give all small stars plus isolated ones. Look at the position of the isolated ones (called here $\backslash\phi_{bis}$) ?

Third inflation with color code from greek letters



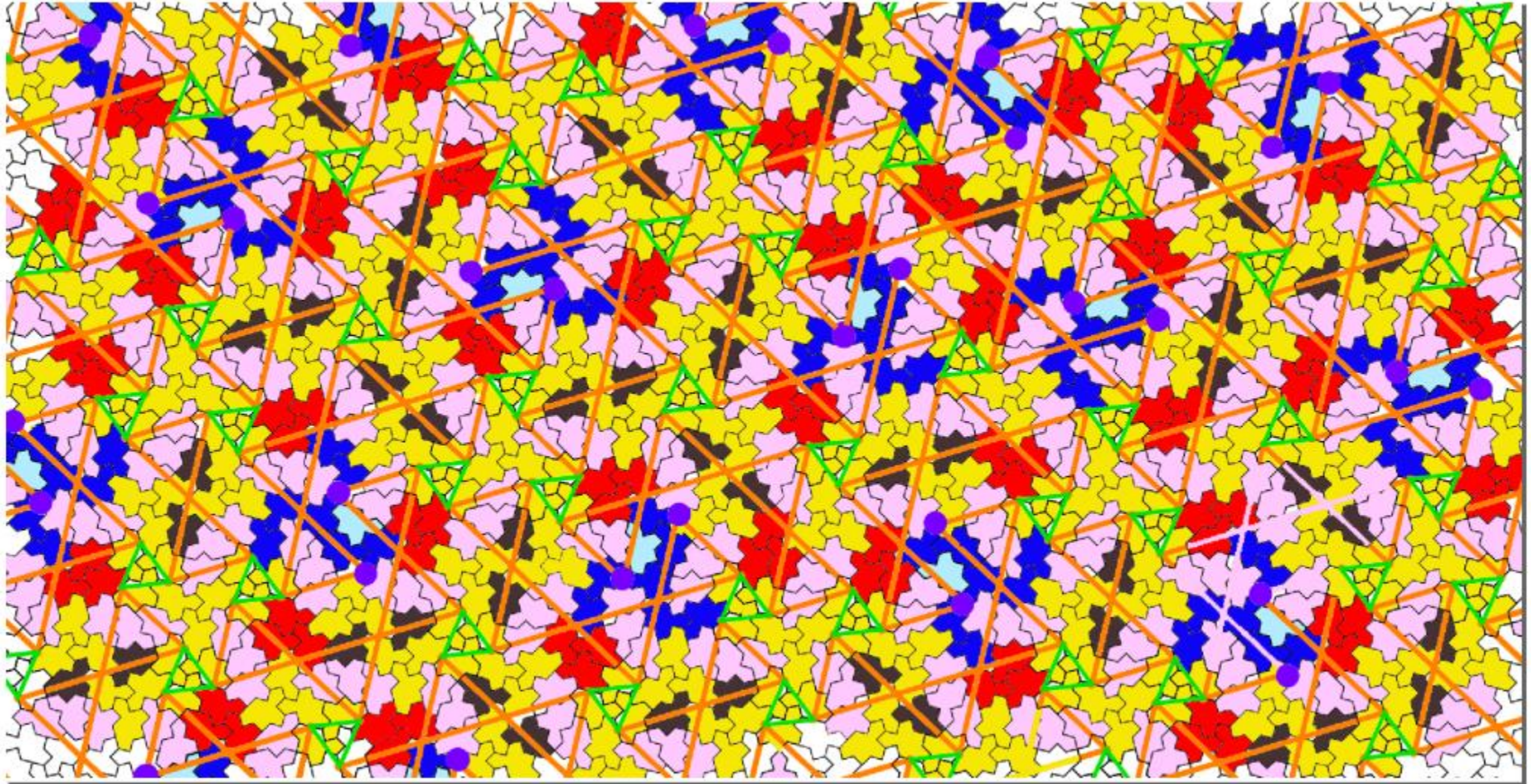
- $\backslash\theta$ and $\backslash\lambda$ always in couple -> part of a propeller
- Large stars correspond to all $\backslash\Pi$ and $\backslash\Psi$ tiles
- $\backslash\phi$ tiles give all small stars plus isolated ones. Look at the position of the isolated ones (called here $\backslash\phi_{bis}$) ?

Another version with less colors ...

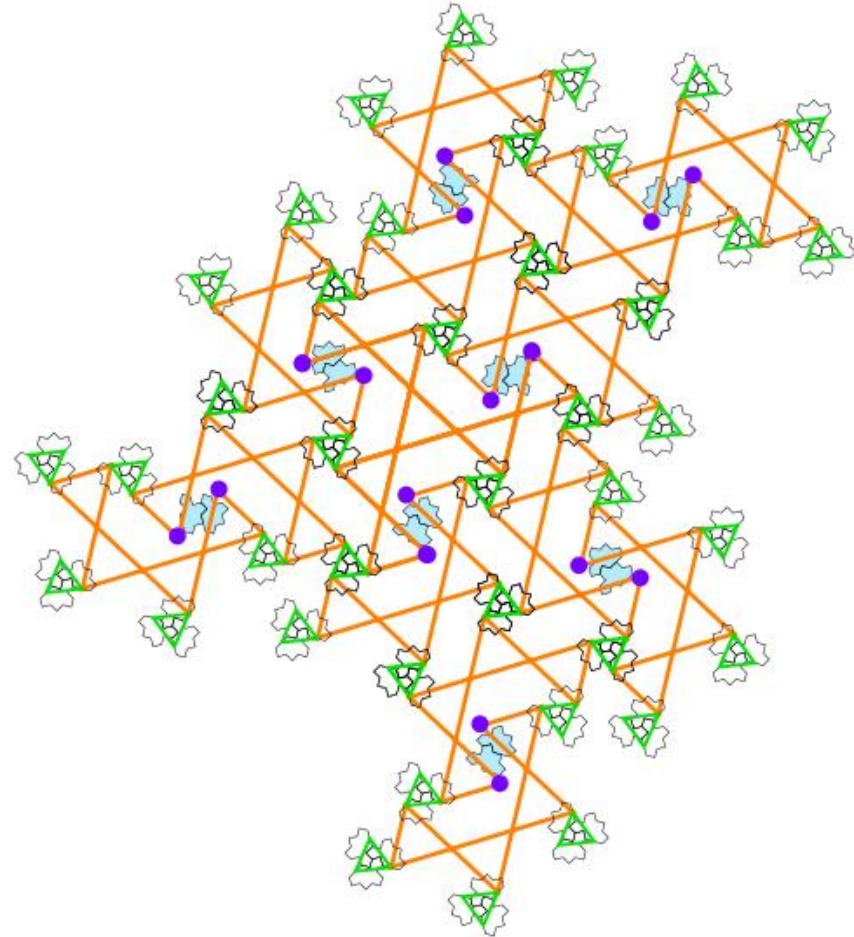
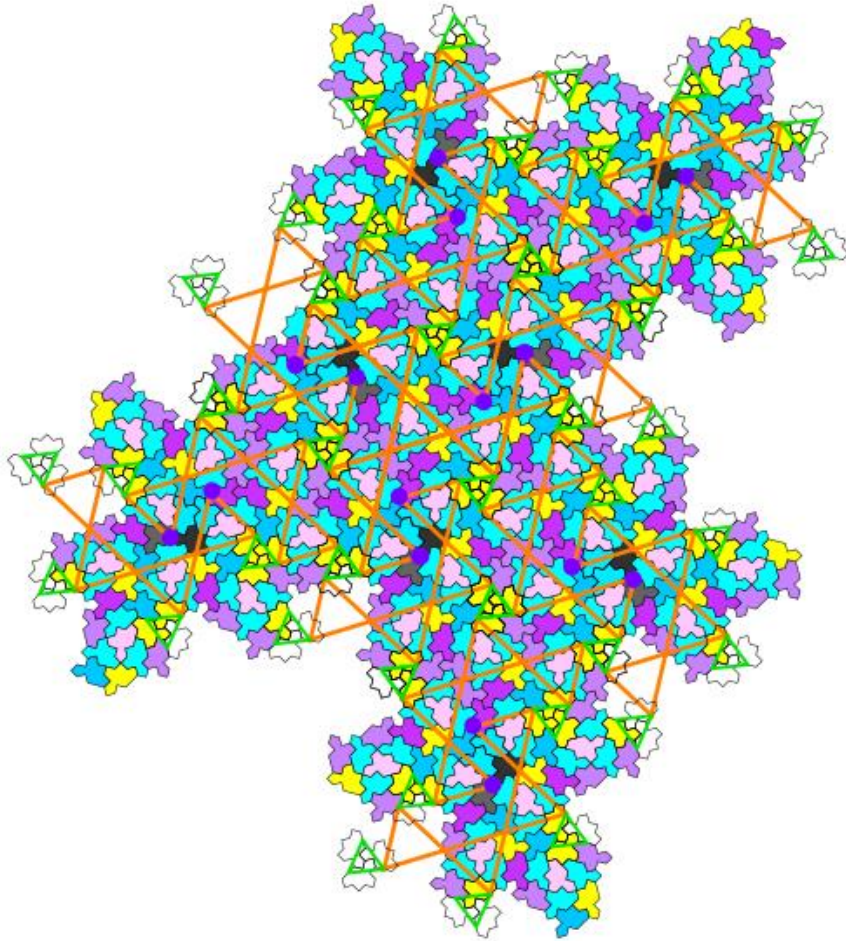


- θ and λ always in couple $\rightarrow$ part of a propeller
- Large stars correspond to all Π and Ψ tiles
- ϕ tiles give all small stars plus isolated ones. Look at the position of the isolated ones (called here ϕ_{bis}) ?

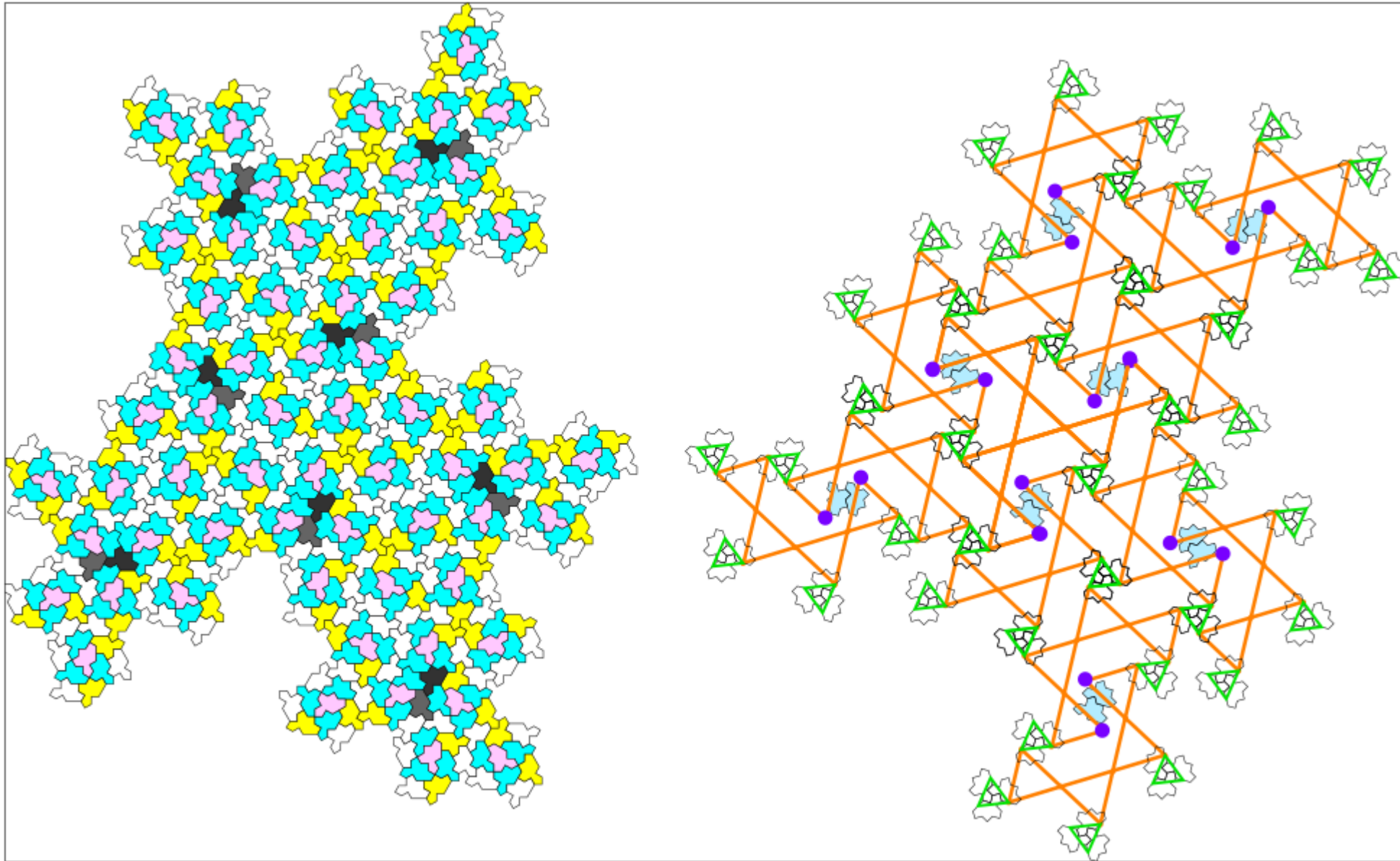
Link with anatomy by Boris Horvat ?



Third inflation: identification of the 2-arms, 3-arms and diabolos

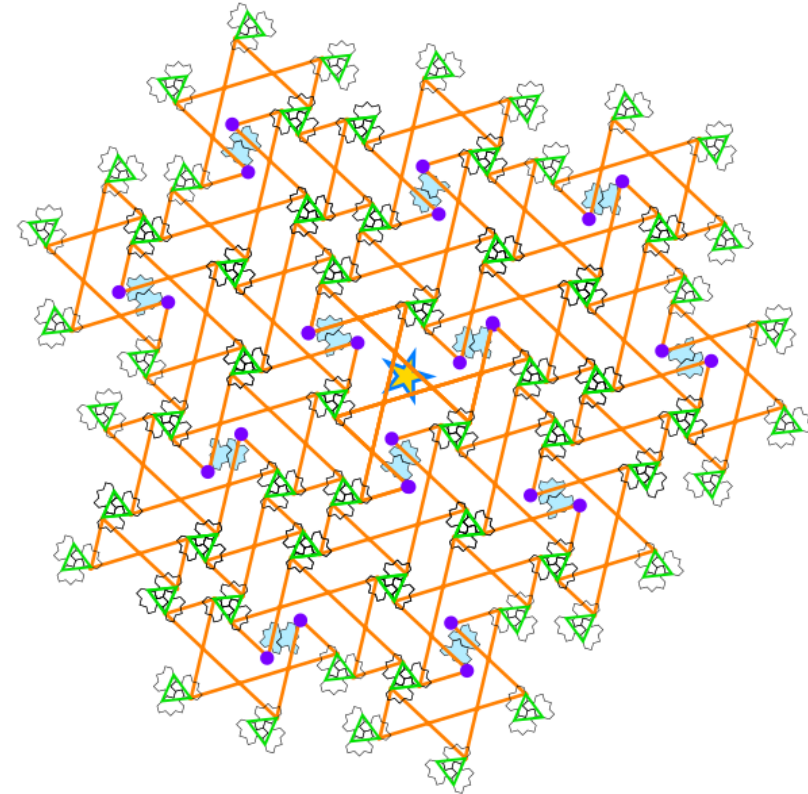
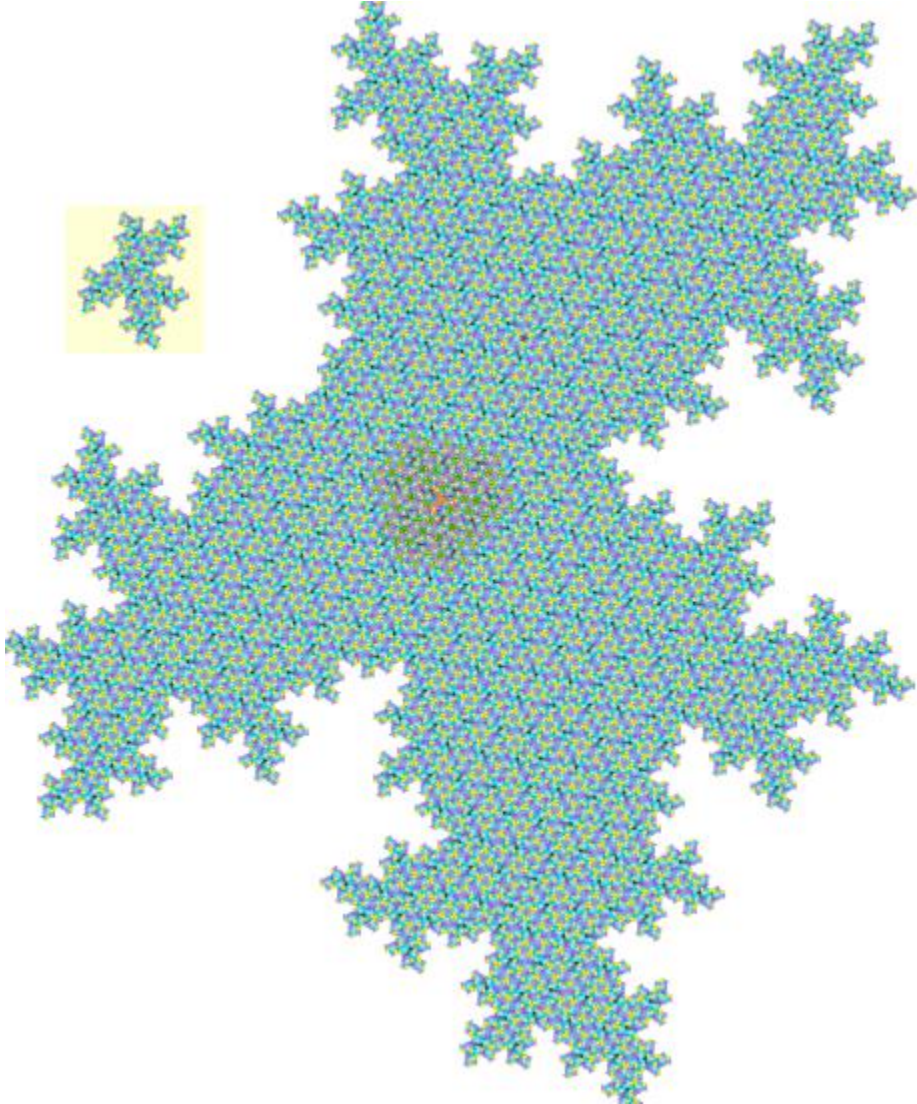


Another version with less colors ...



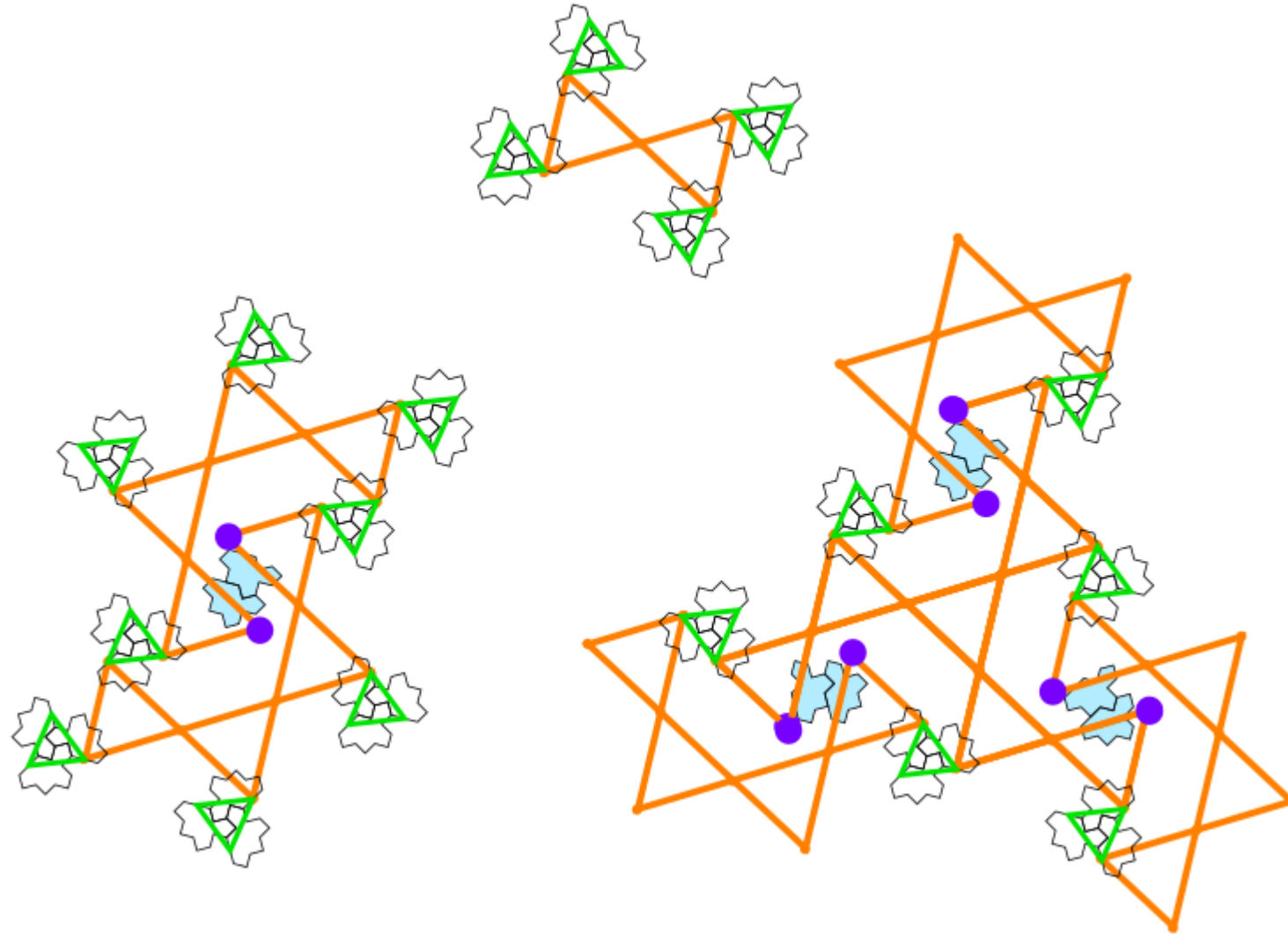
■ $\backslash\theta$ and $\backslash\lambda$ couple
-> part of a propeller ->
highlight a specific
configuration
of two full
clusters !

Third to fifth inflation: identification of the 2-arms, 3-arms and diabolos ...

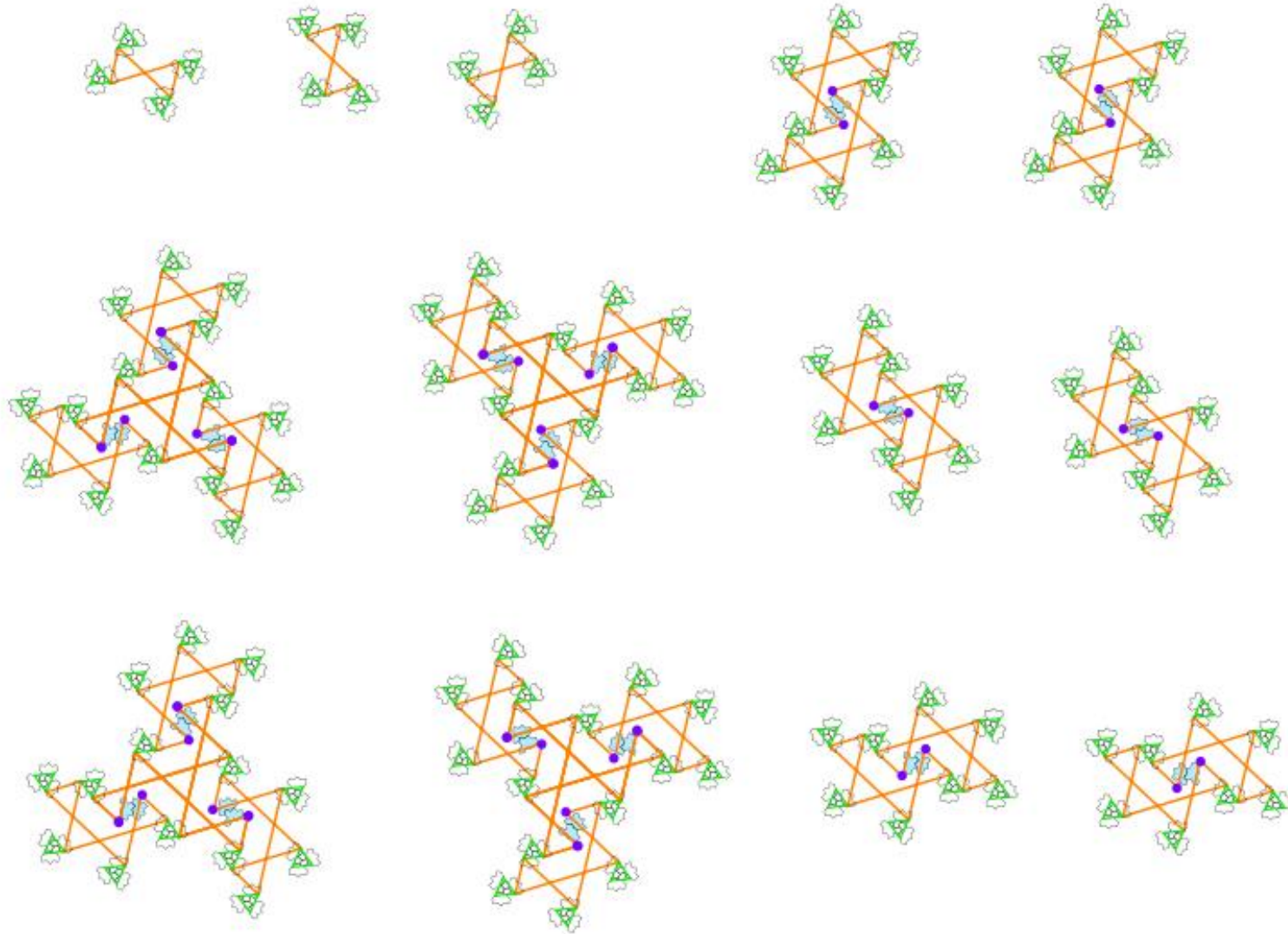


- Very long to do by hand !!!

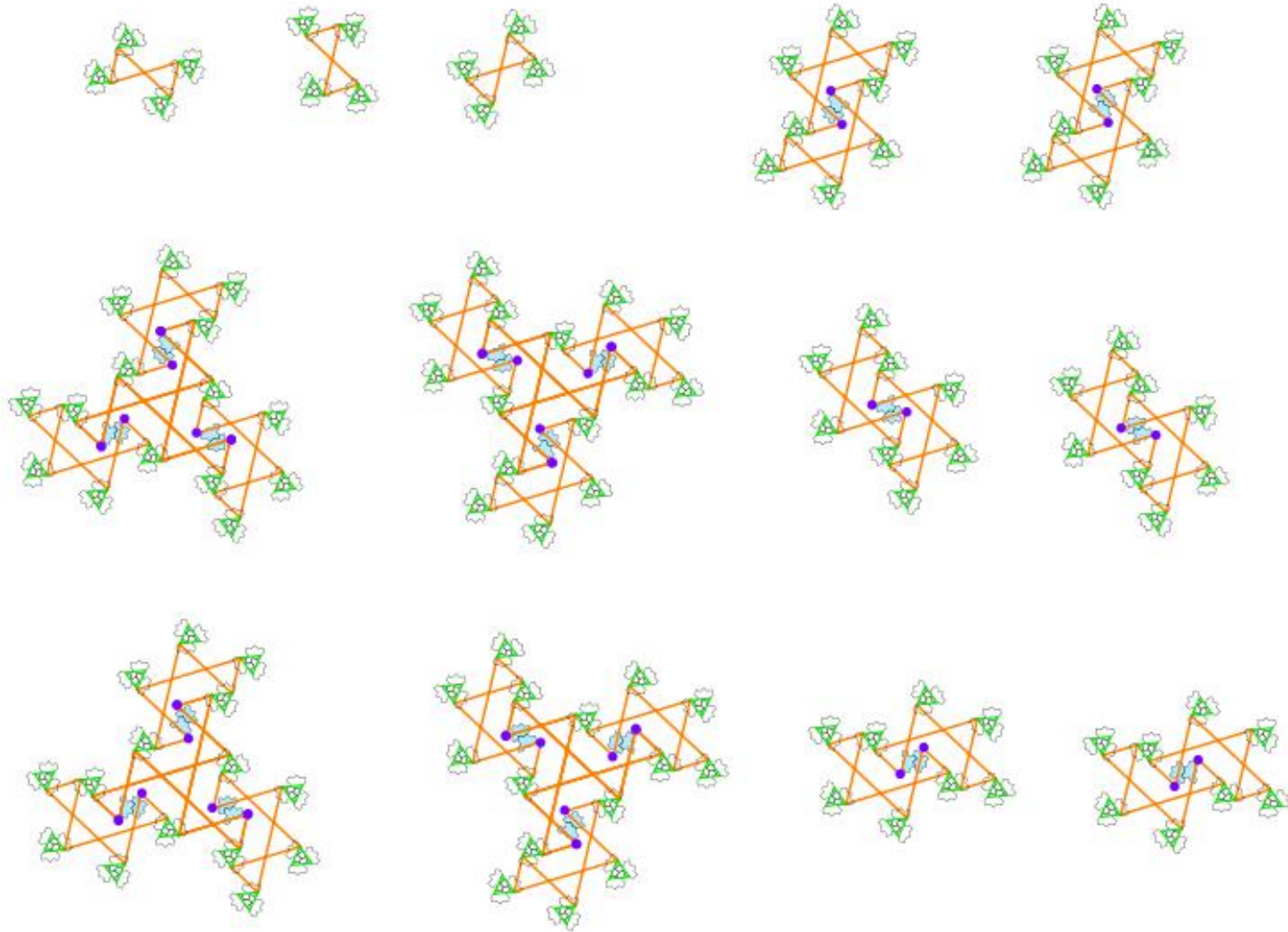
New puzzle



New puzzle

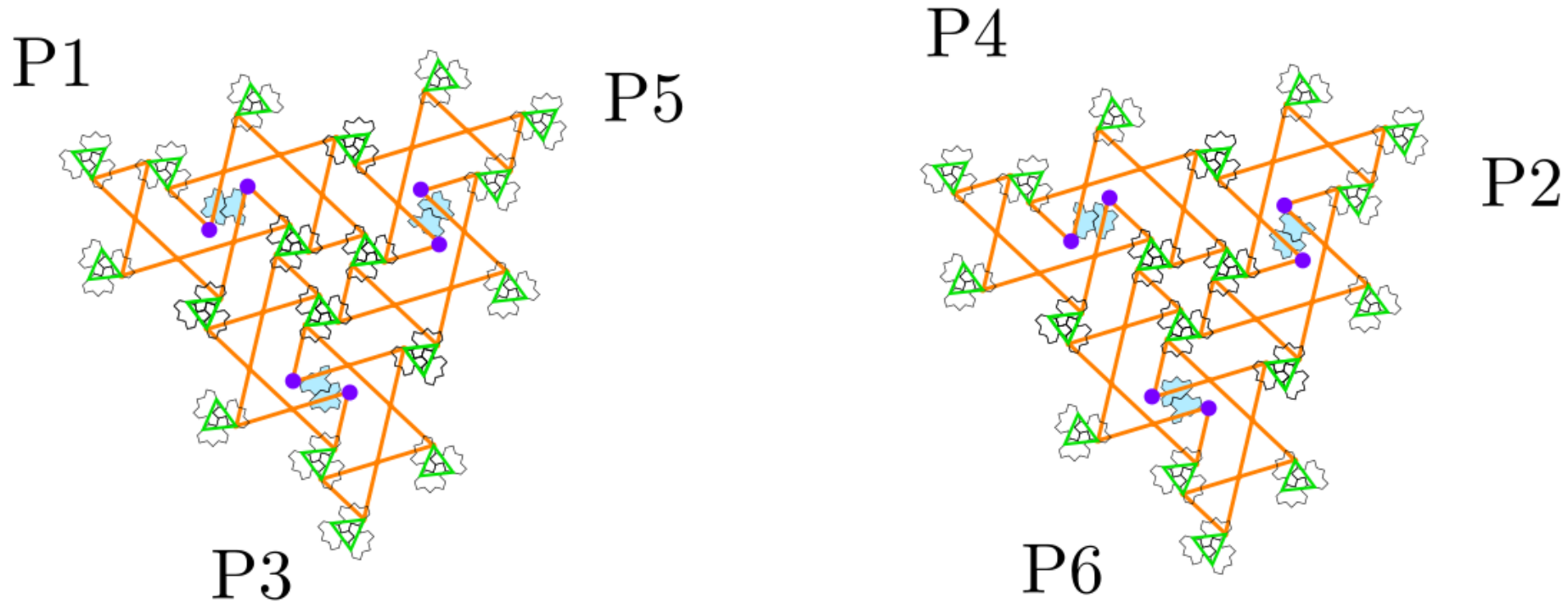


Homochiral assembly rule from these pieces ???



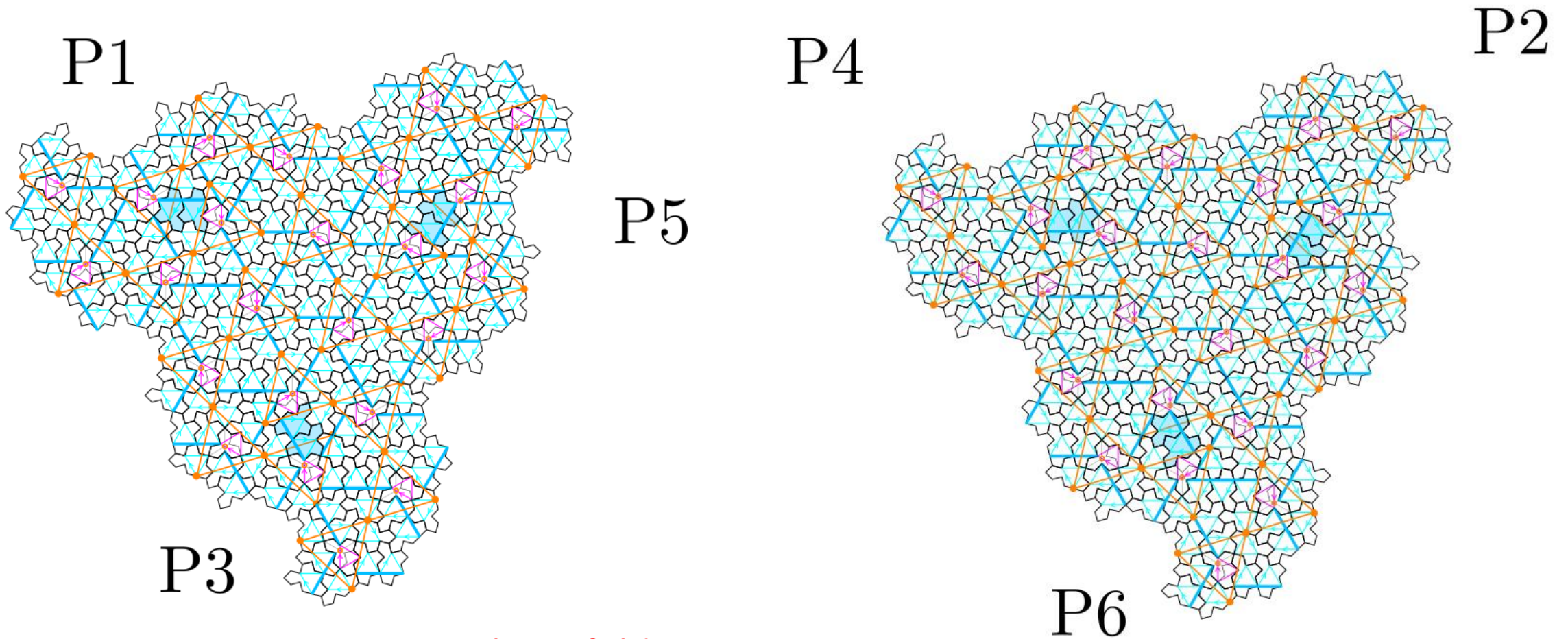
Possible candidate for a seed ?

- Three 2-arms combined in this way:



Possible candidate for 3-fold symmetry center

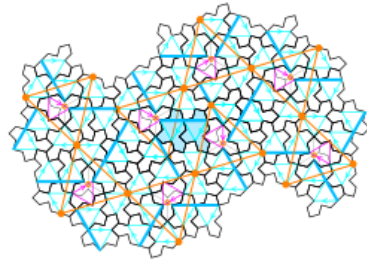
- Compare with center of giant star cluster by Boris Horvat ???



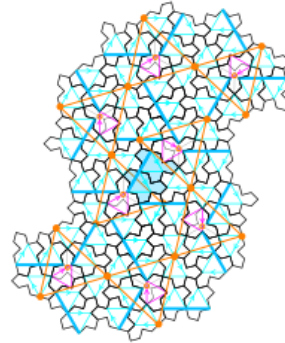
- True three fold symmetry point at center ???

Six orientations of the 2-arms

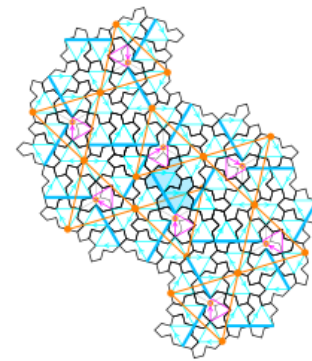
P1



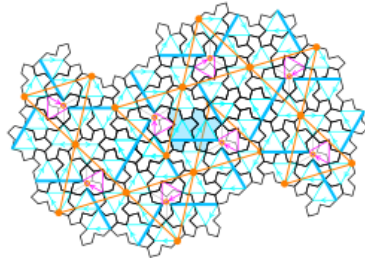
P2



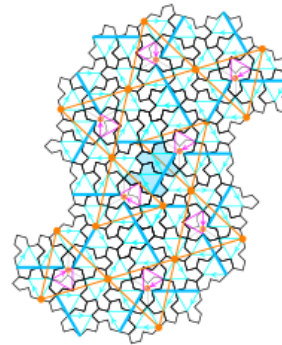
P3



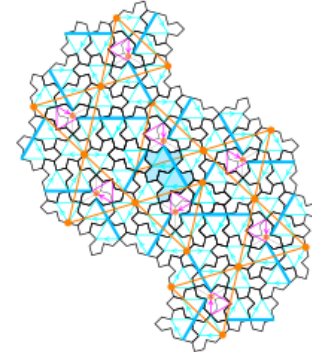
P4



P5

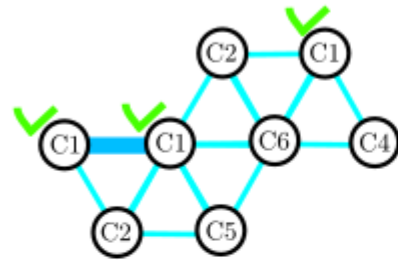


P6

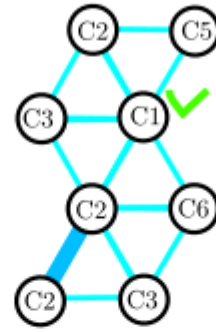


Six orientations of the 2-arms

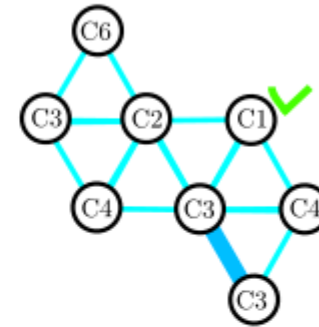
P1



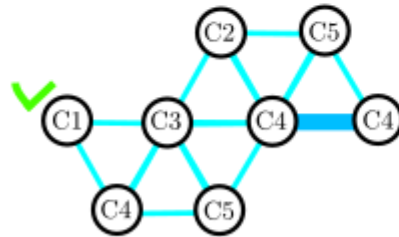
P2



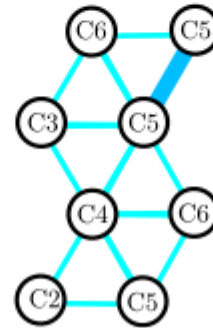
P3



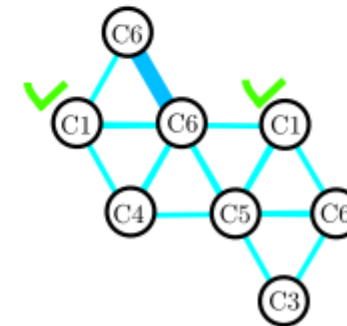
P4



P5



P6

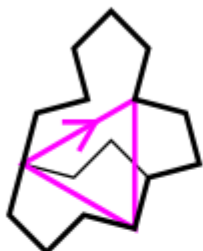


ODD and EVEN orientations for Tile(1,1)

$M_1 = M_{O_1}$



$M_3 = M_{O_2}$



$M_5 = M_{O_3}$



$M_7 = M_{O_4}$



$M_9 = M_{O_5}$



$M_{11} = M_{O_6}$



$M_2 = M_{E_1}$



$M_4 = M_{E_2}$



$M_6 = M_{E_3}$



$M_8 = M_{E_4}$

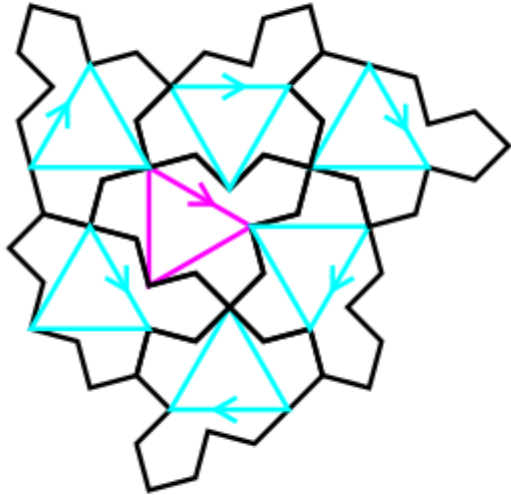


$M_{10} = M_{E_5}$

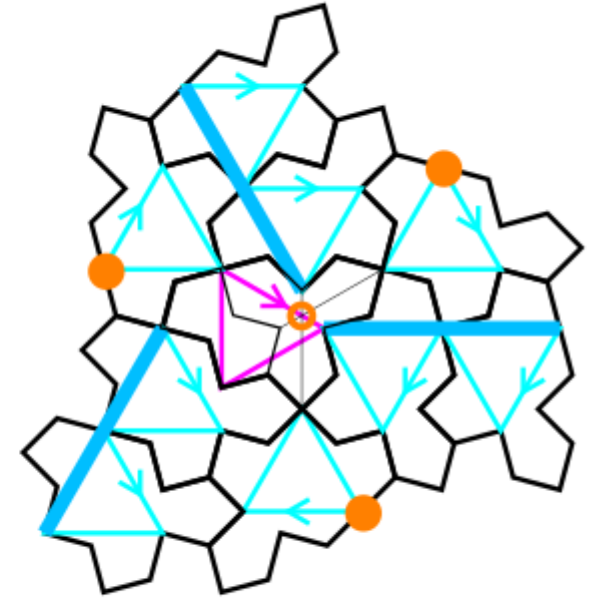
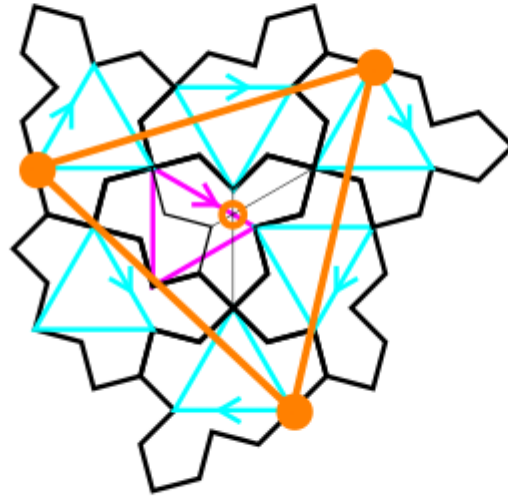


$M_{12} = M_{E_6}$

- Tile(1,1)



full cluster

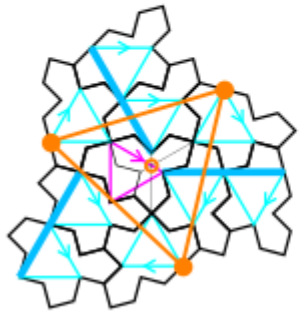


- C cluster: one ODD tile at the center + 6 EVEN tiles around
- Three glue points (orange color): regular triangle centered on the trifle shape of the ODD tile
- Full cluster: 3 more EVEN tiles obtained by translations (three dark blue lines)

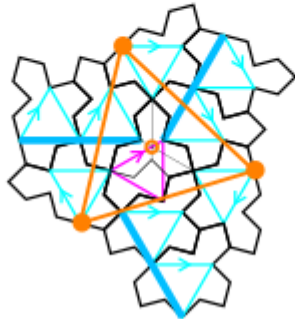
full cluster with its six rotations: $n(2\pi/6)$

- Tile(1,1)

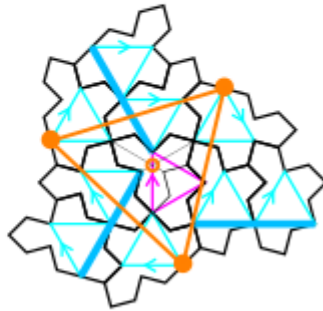
C1



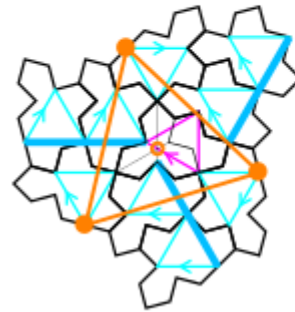
C2



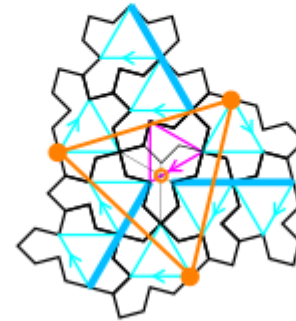
C3



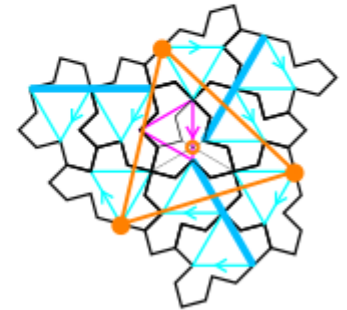
C4



C5

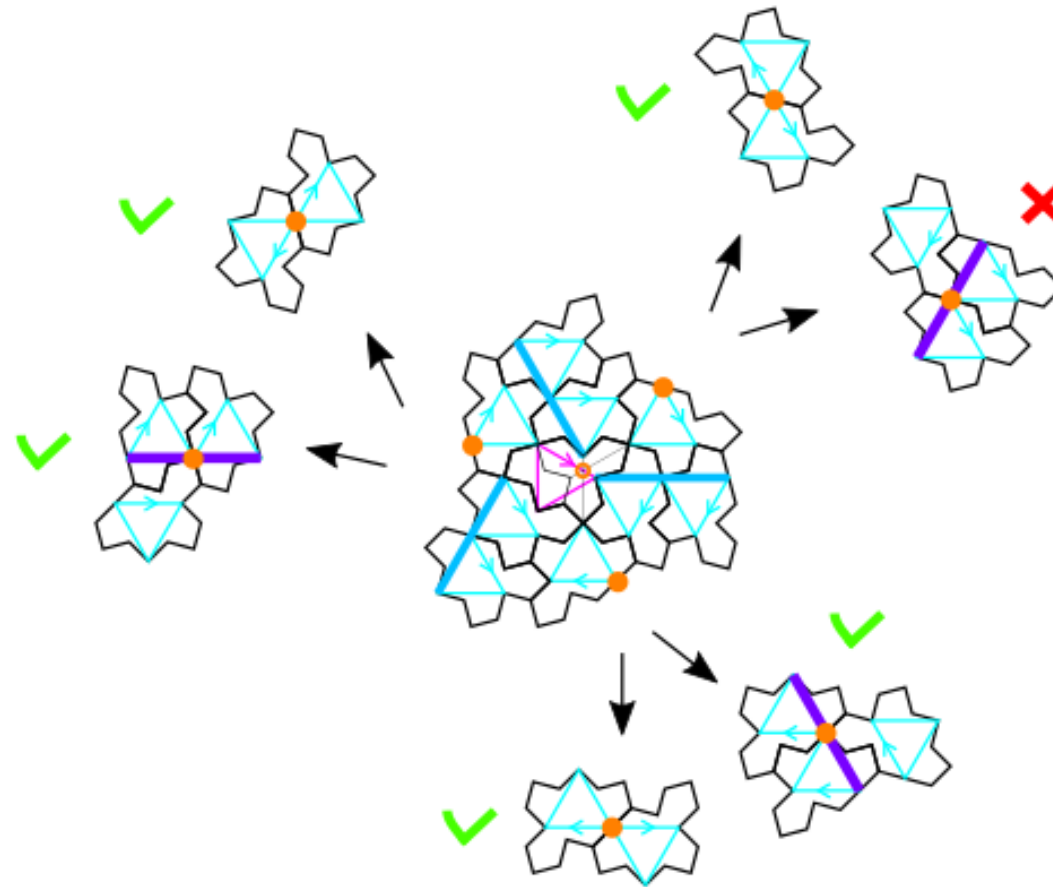


C6



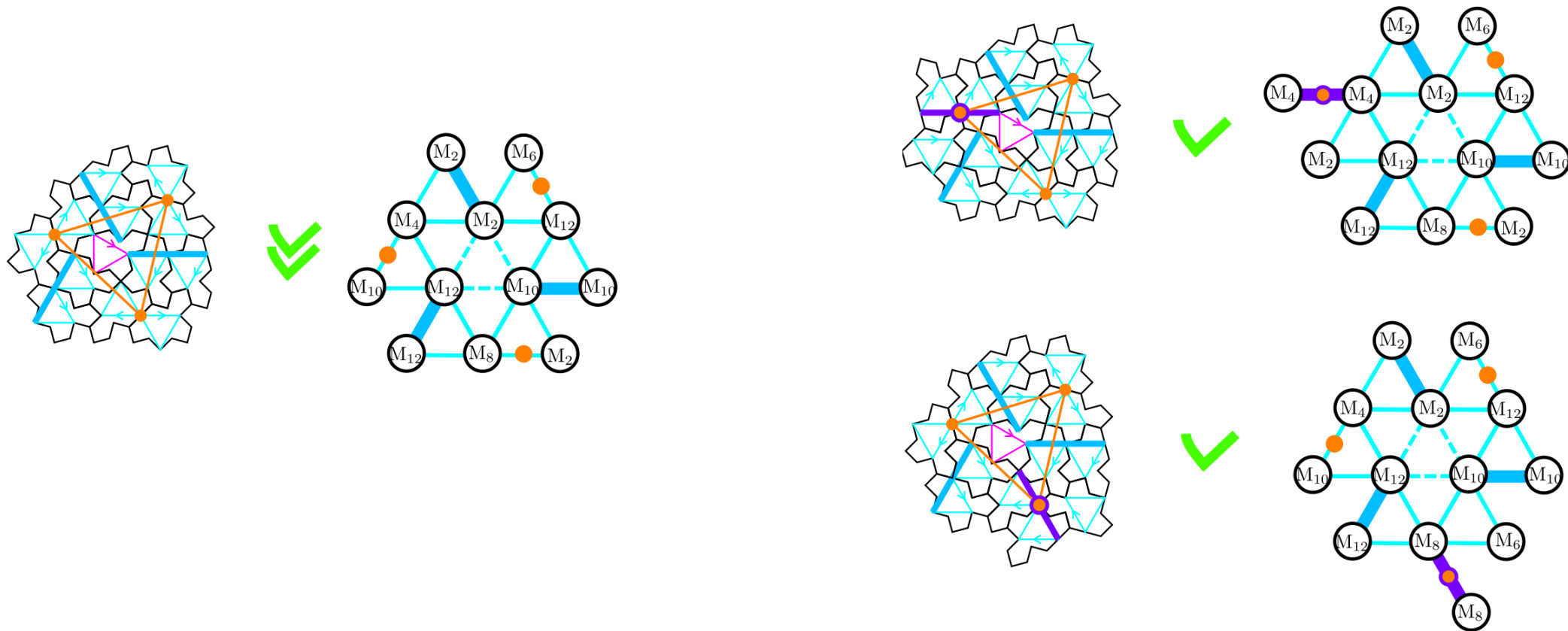
- 13 tiles in a full cluster: one ODD tile at the center + 9 EVEN tiles around
- Blue triangles (EVEN) have one horizontal edge when pink (ODD) have a vertical one

Possible configurations at the glue points ?



- Half-turned or translated monotile (violet line)

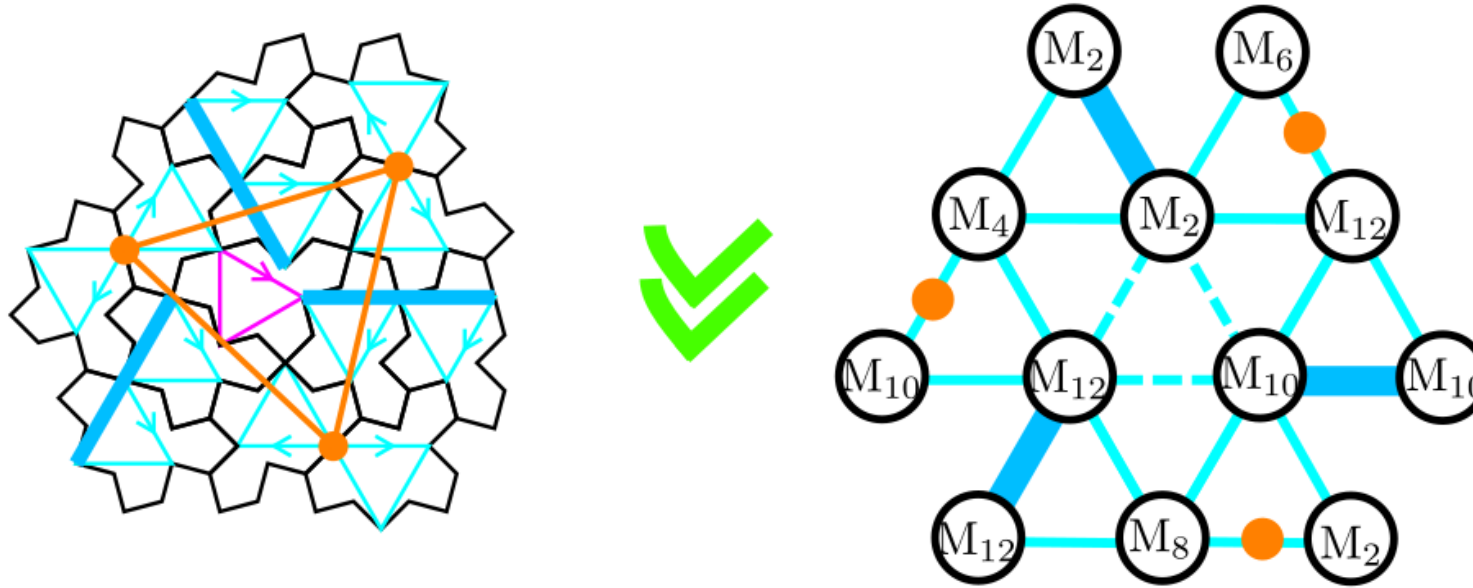
Three possible configurations at the glue points



- configuration with three half-turned tiles is the most common
- The other two are always coupled together in the tiling

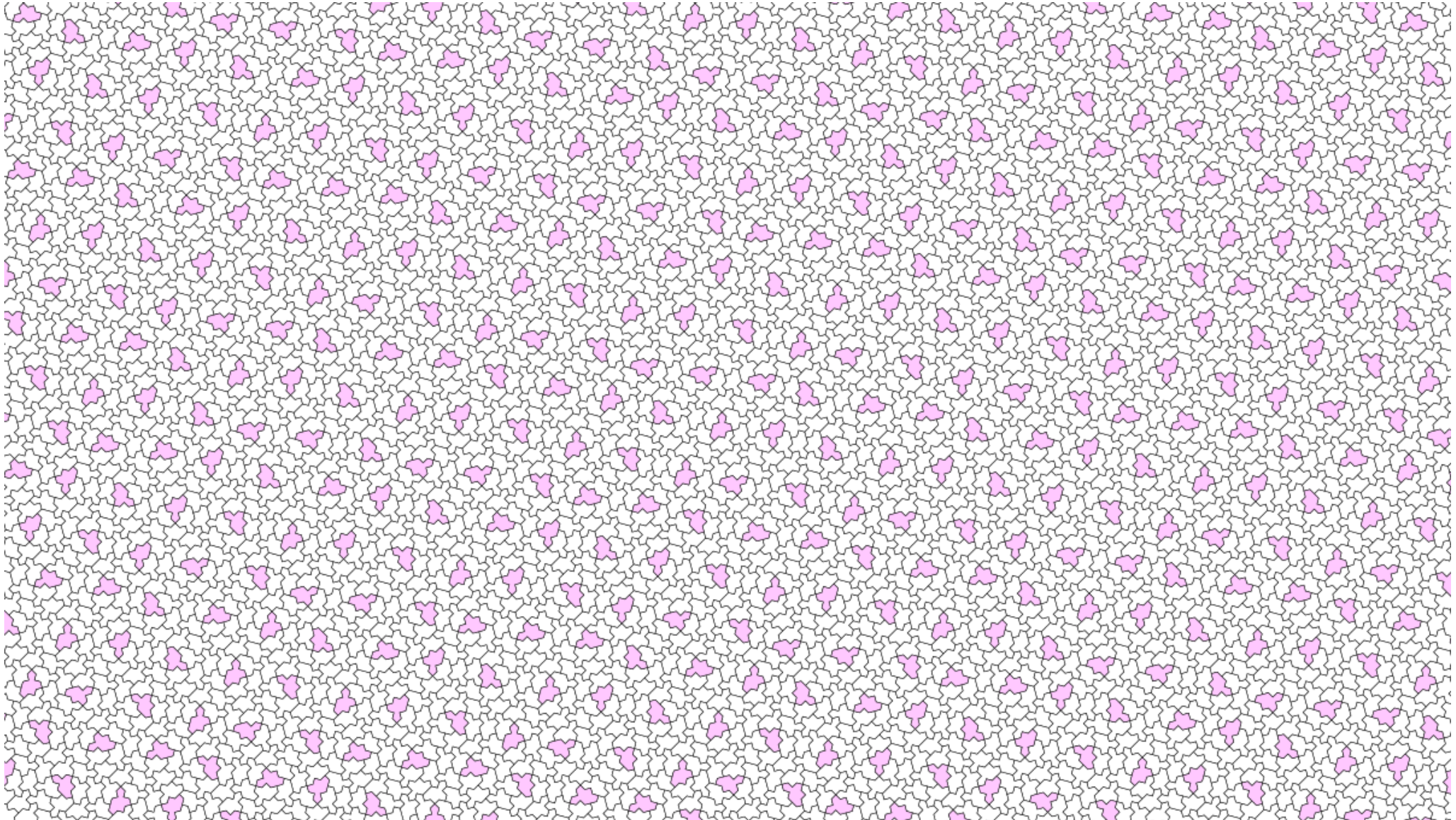
Orientation of the EVEN tiles in the most frequent configuration

- Tile(1,1)



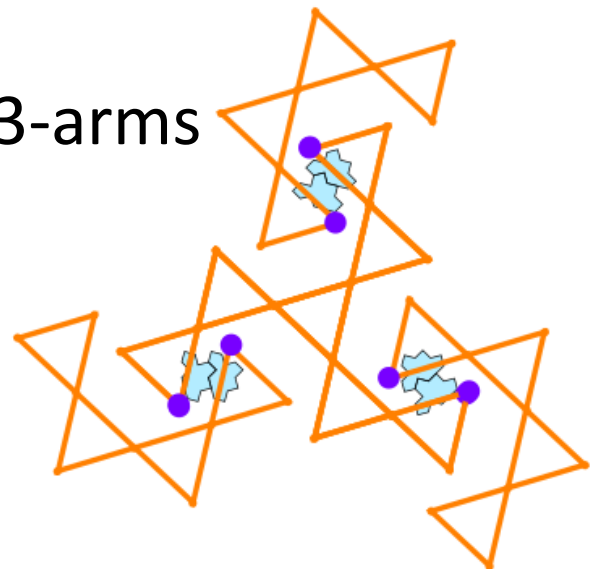
- 6 tiles in the C-cluster (corona made of M12, M4, M2, M12, M10, M8)
- 3 translated (M2, M10, M12) (along blue lines, the edges of blue triangles)
- 3 half-turned (M4 → M10, M12 → M6, M8 → M2)

Construction for the whole tiling ???

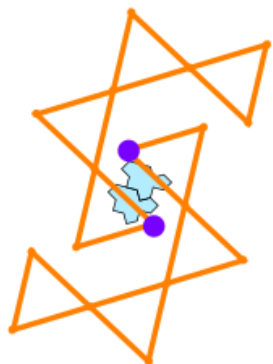


Three patterns in the whole tiling 😊

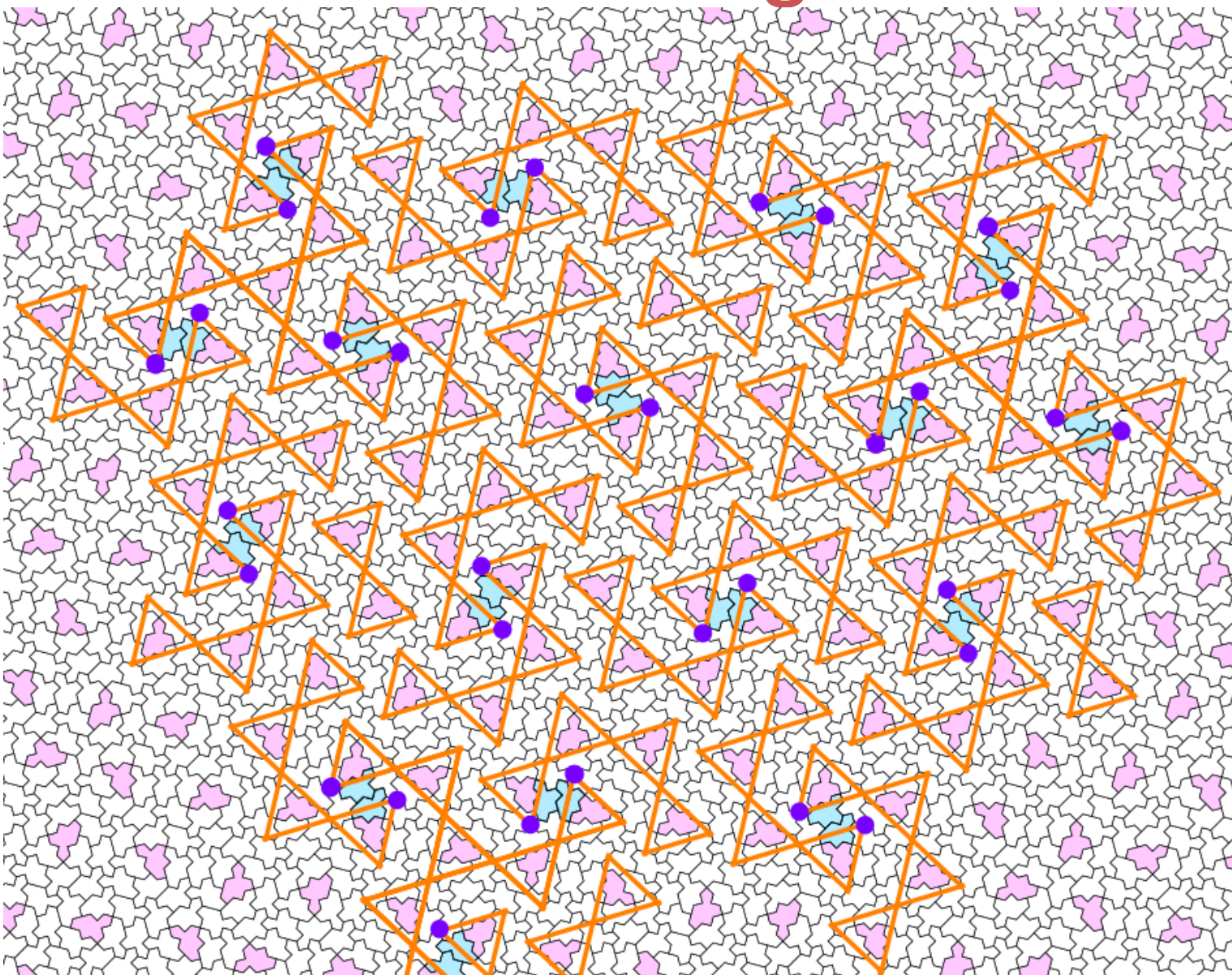
■ 3-arms



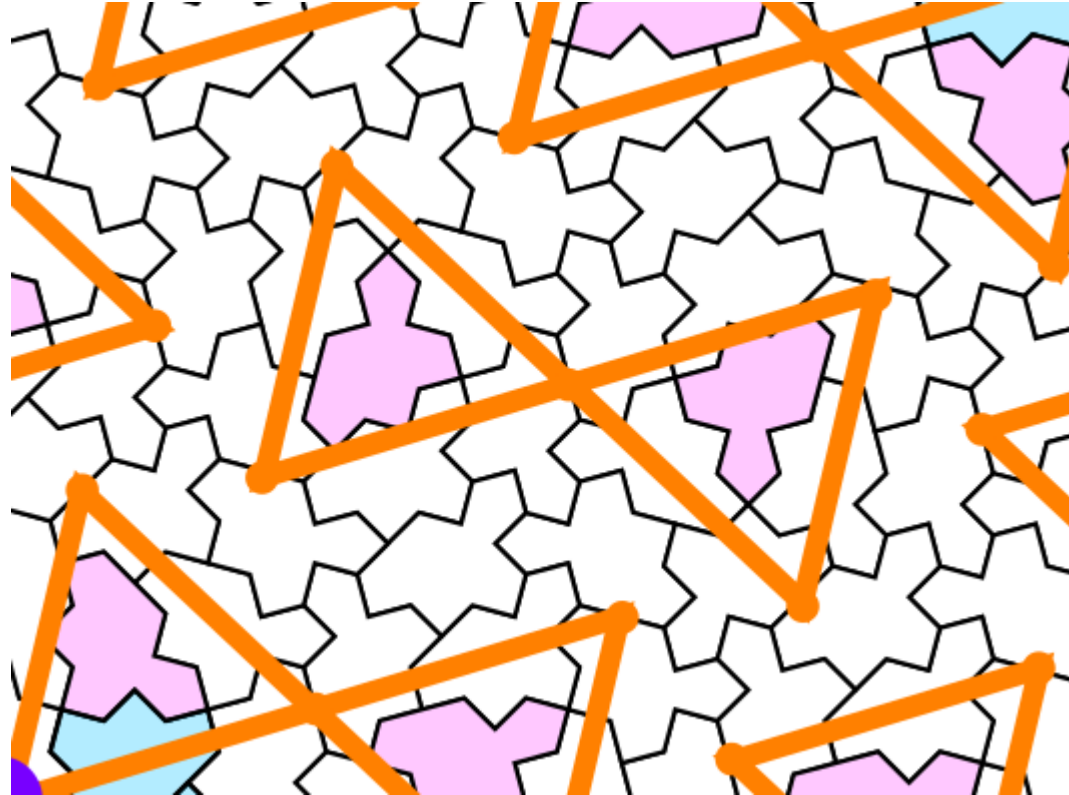
■ 2-arms



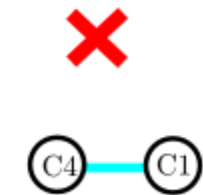
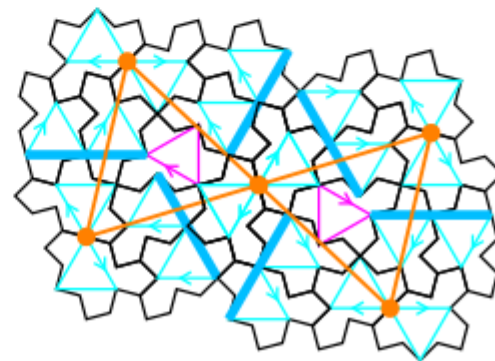
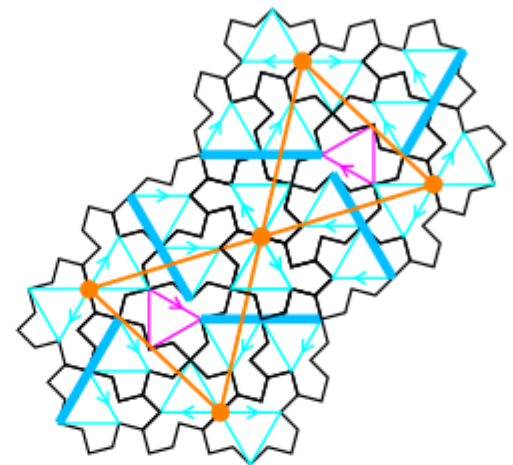
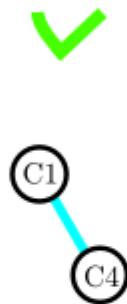
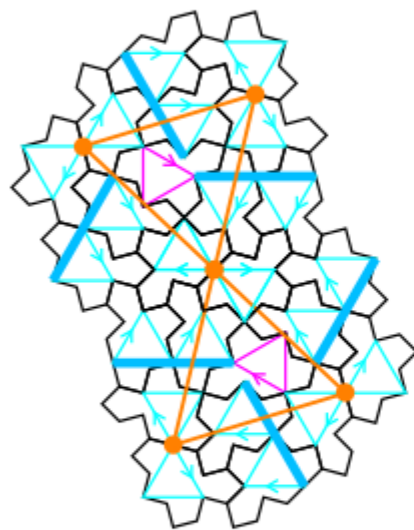
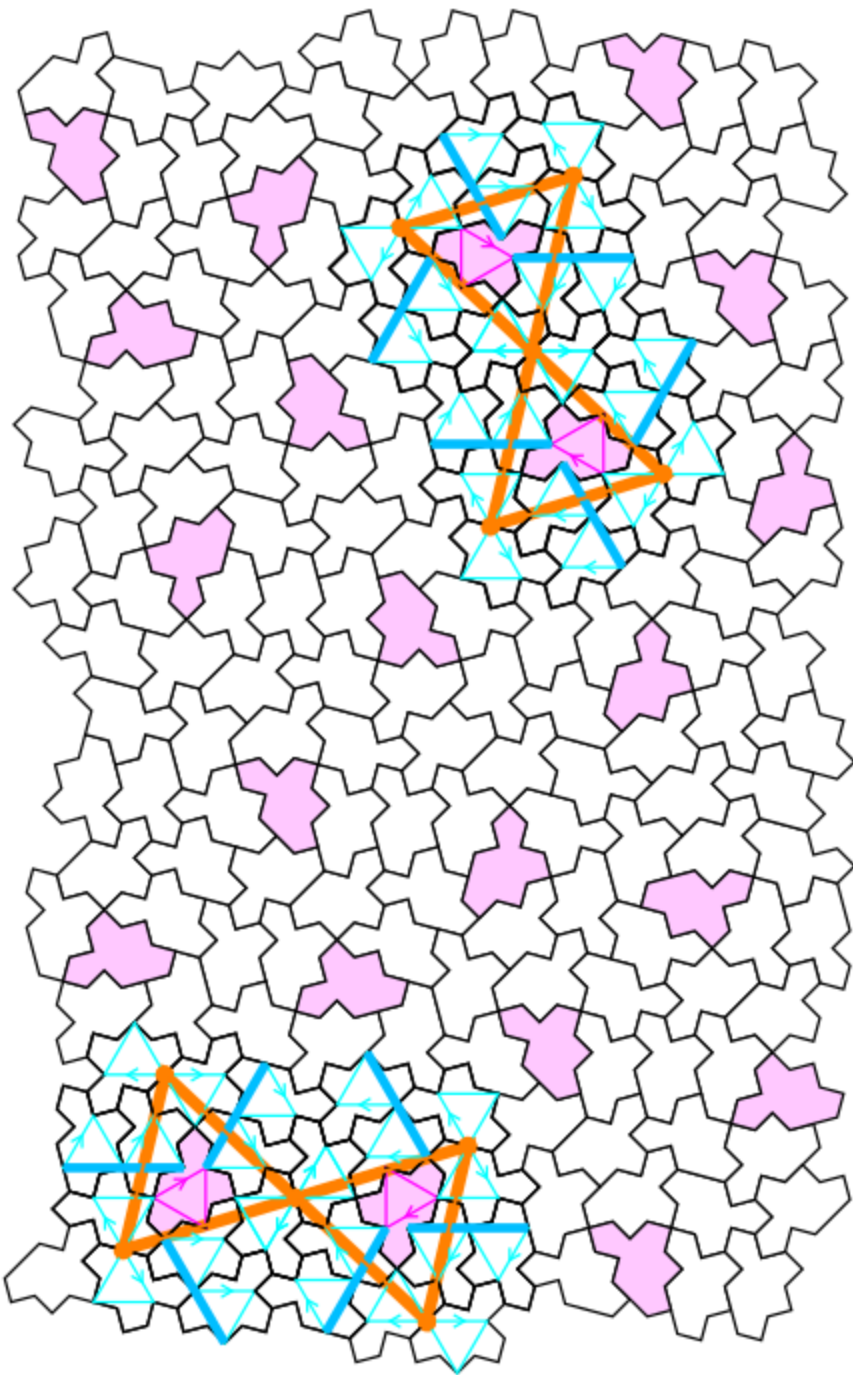
■ diabolos



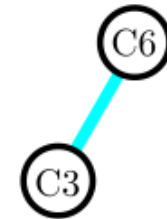
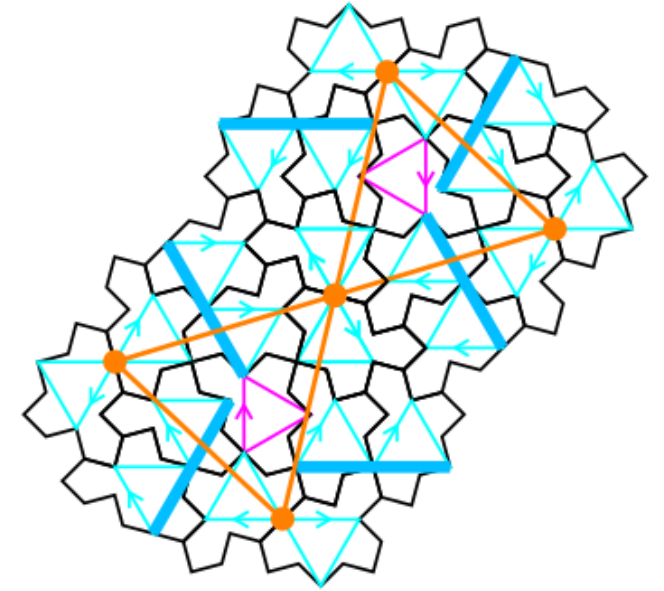
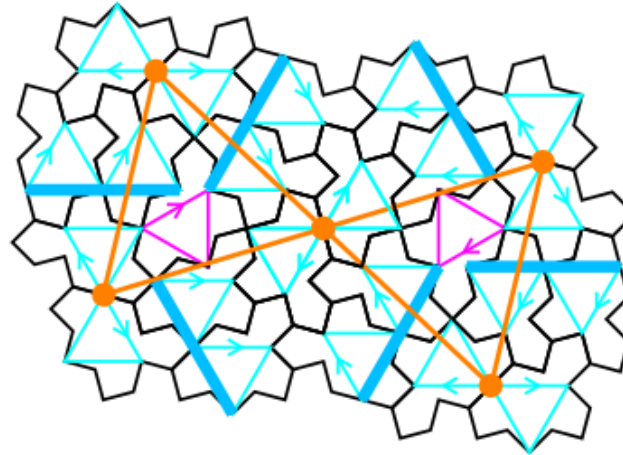
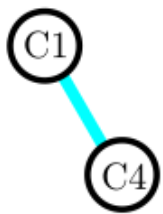
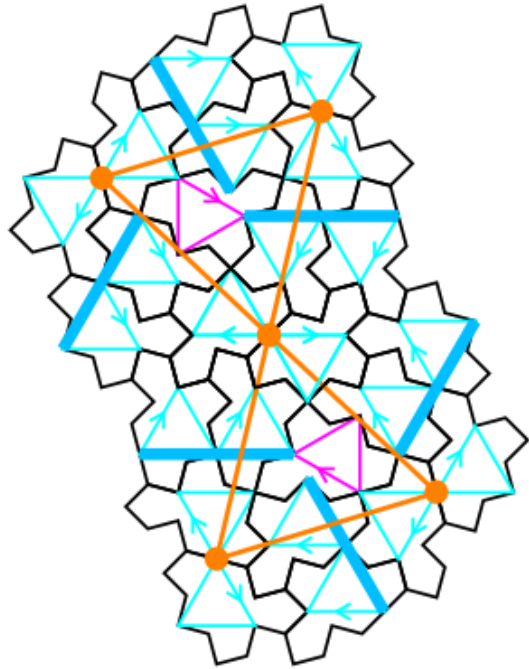
Diabolo pattern



Diabolo pattern

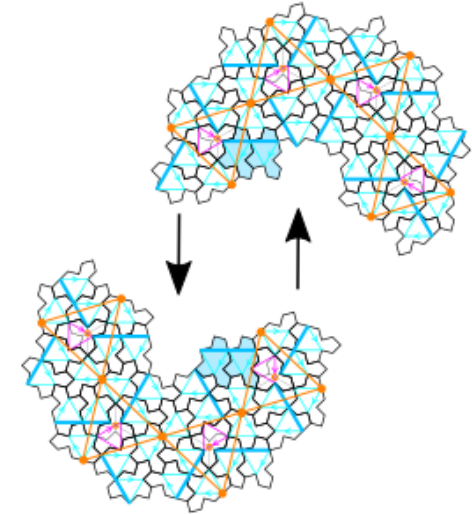
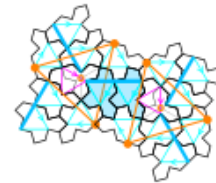
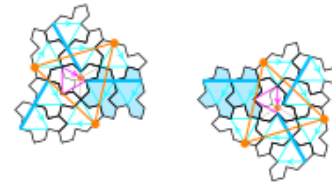
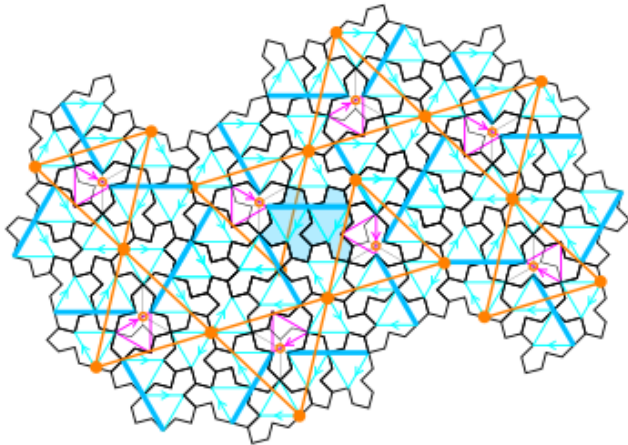
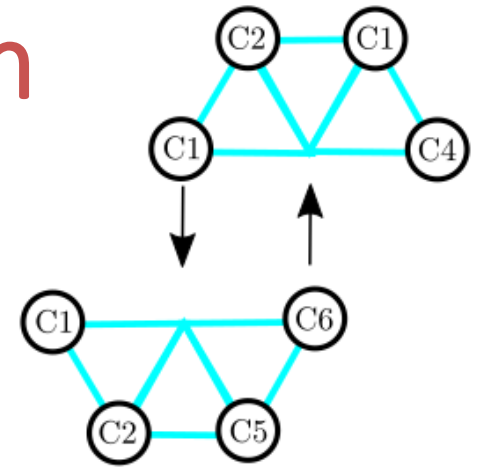
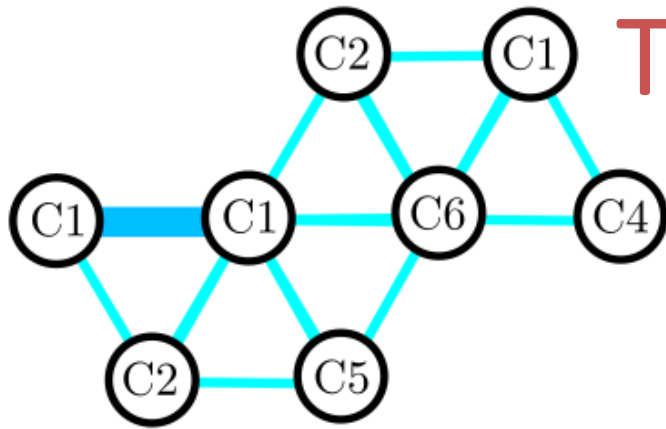


Diabolo pattern



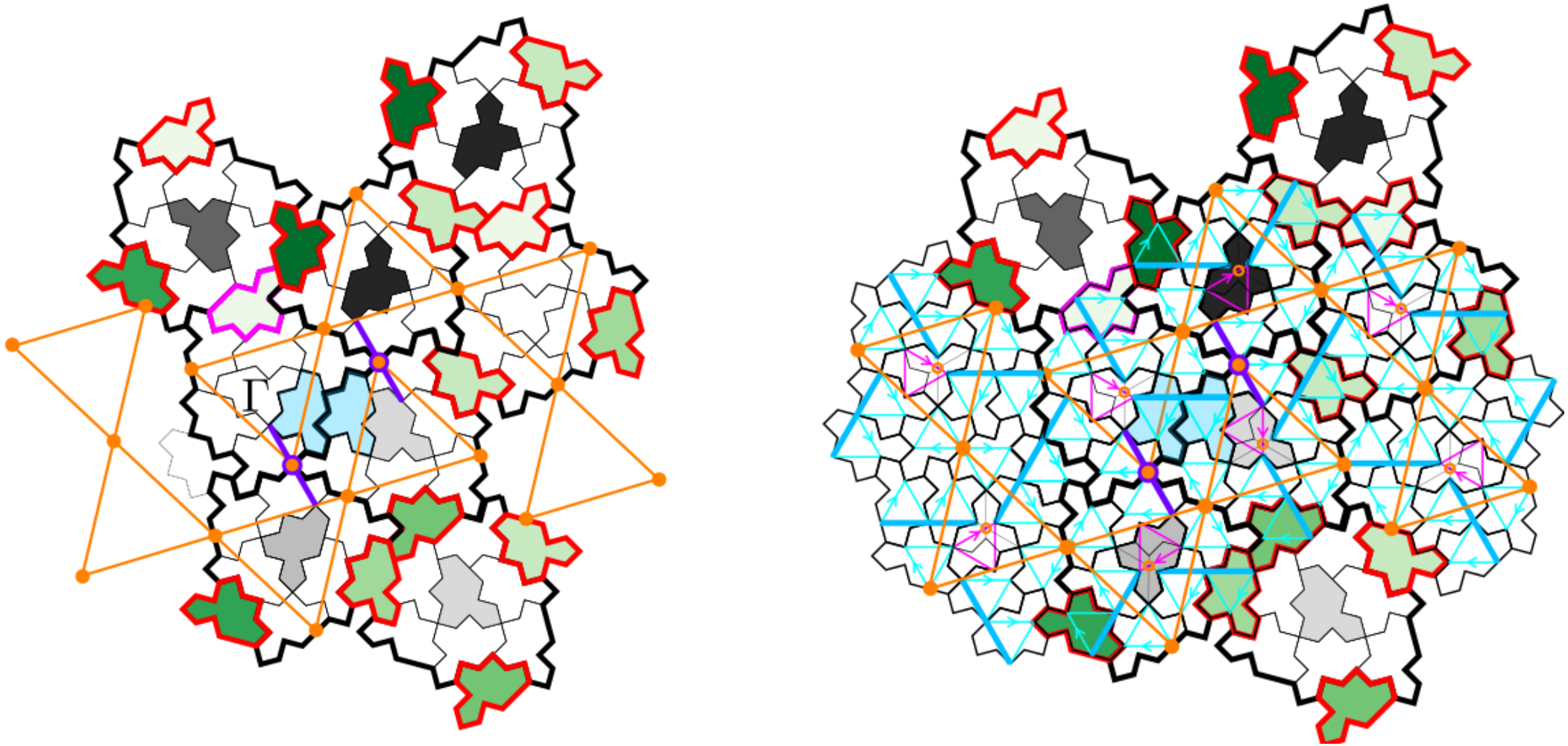
- Three orientations of the same diabolo pattern

Two-arms chiral pattern



- Two 'croissants' of 4 clusters
- Two translated monotiles in the middle + superposition of glue points
- Easy to code ! 😊

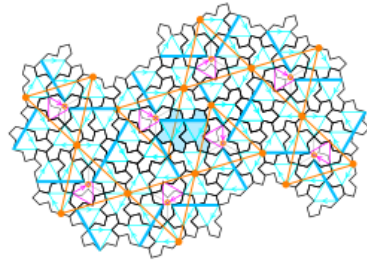
Link with triangular pattern



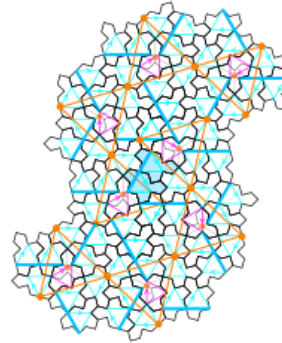
- The two other environnements with a violet lines are always coupled together

Six orientations of the 2-arms

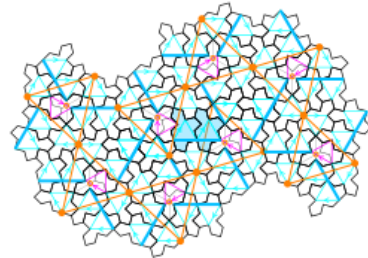
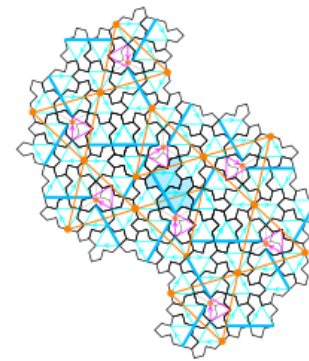
P1



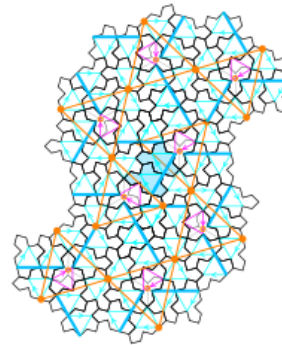
P2



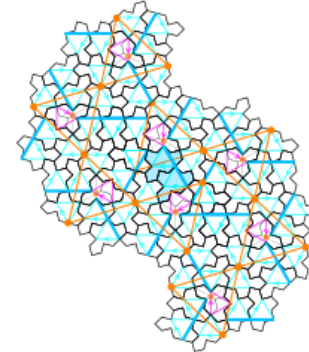
P3



P4



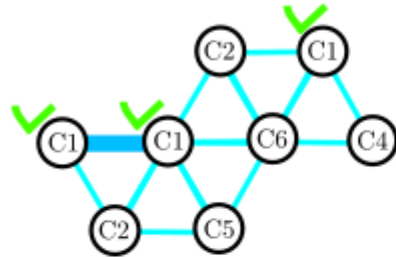
P5



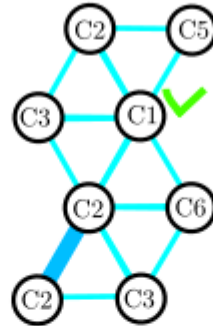
P6

Six orientations of the 2-arms

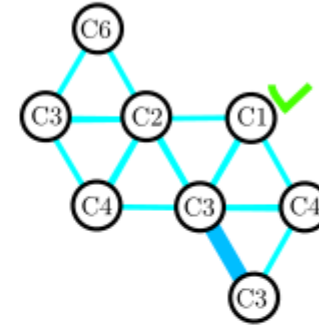
P1



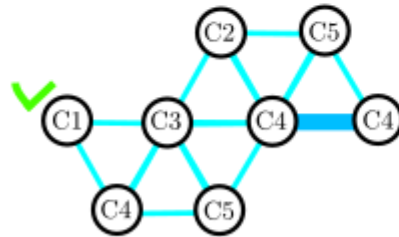
P2



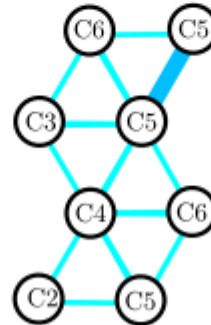
P3



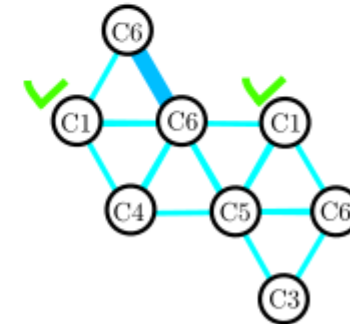
P4



P5

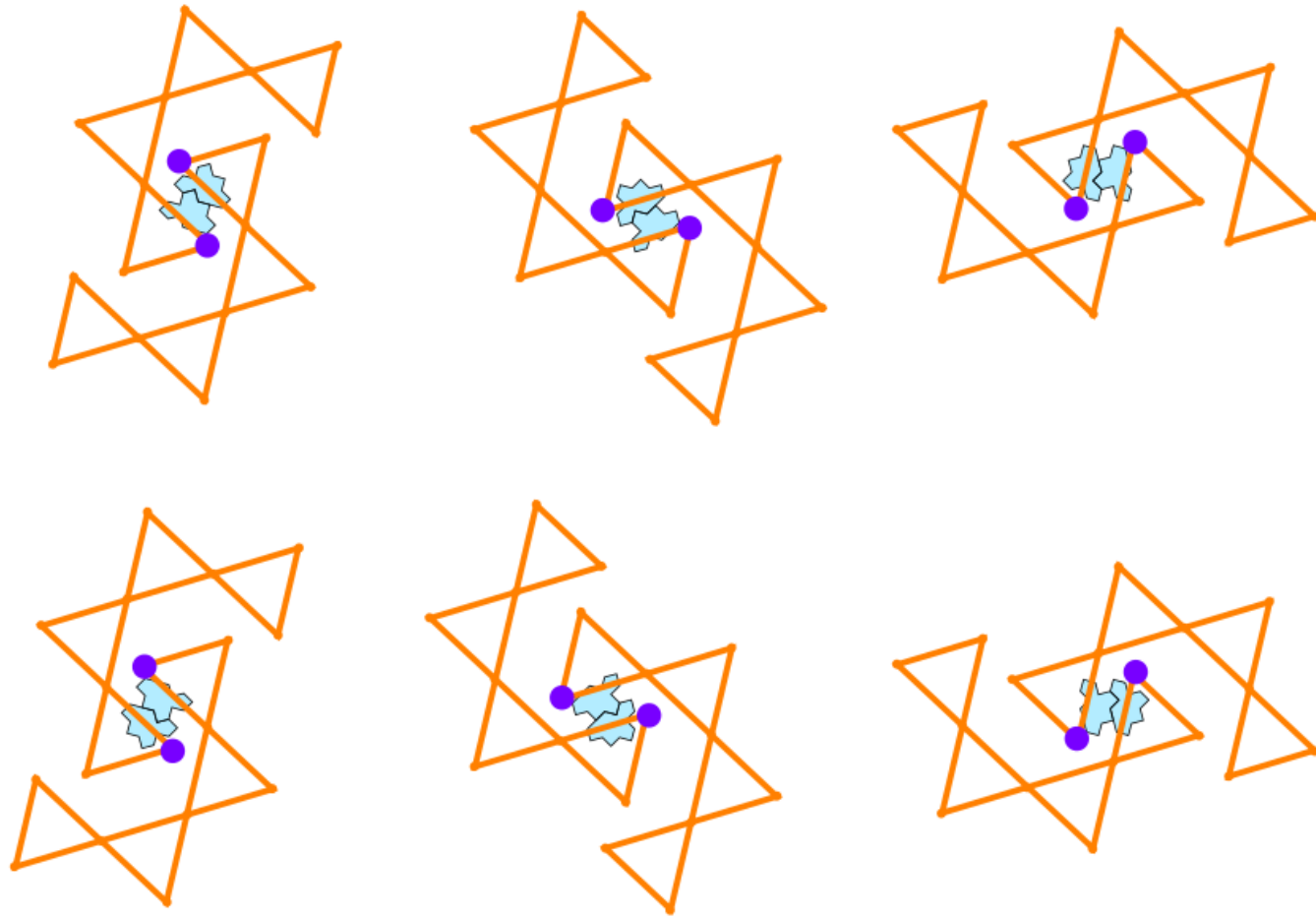


P6

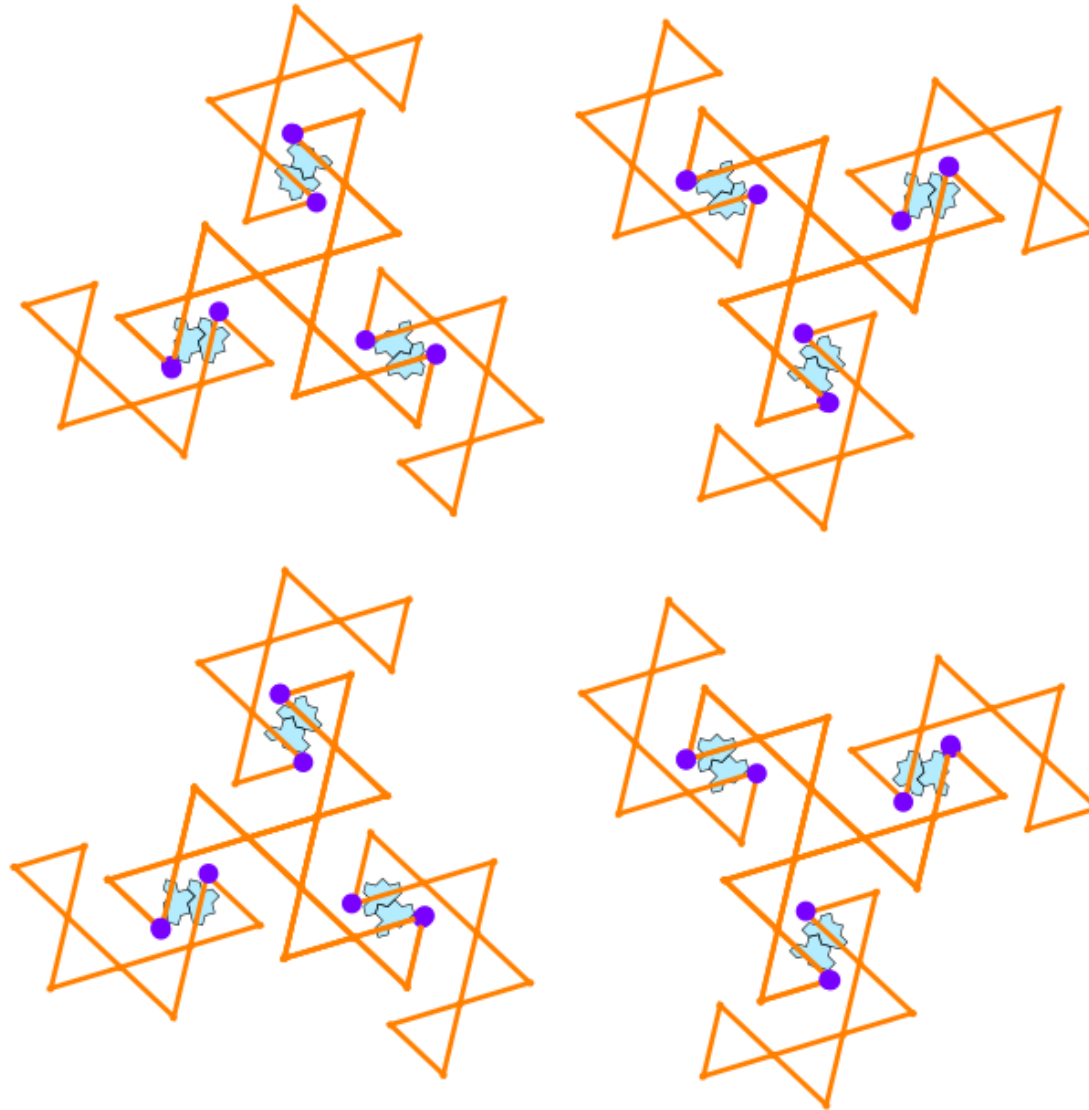


- 8 combinations for a given cluster orientation (green crosses)

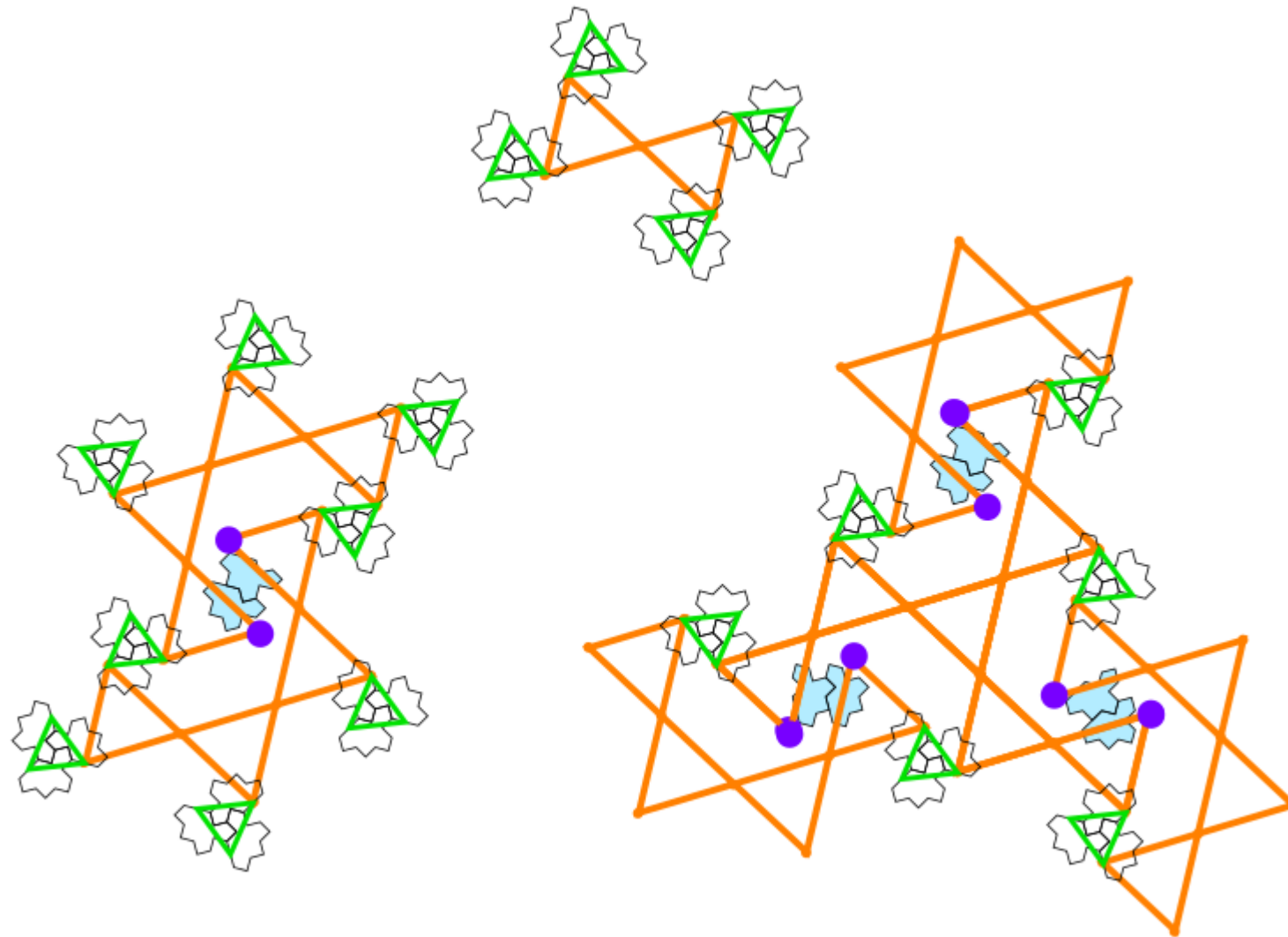
Six orientations of the 2-arms



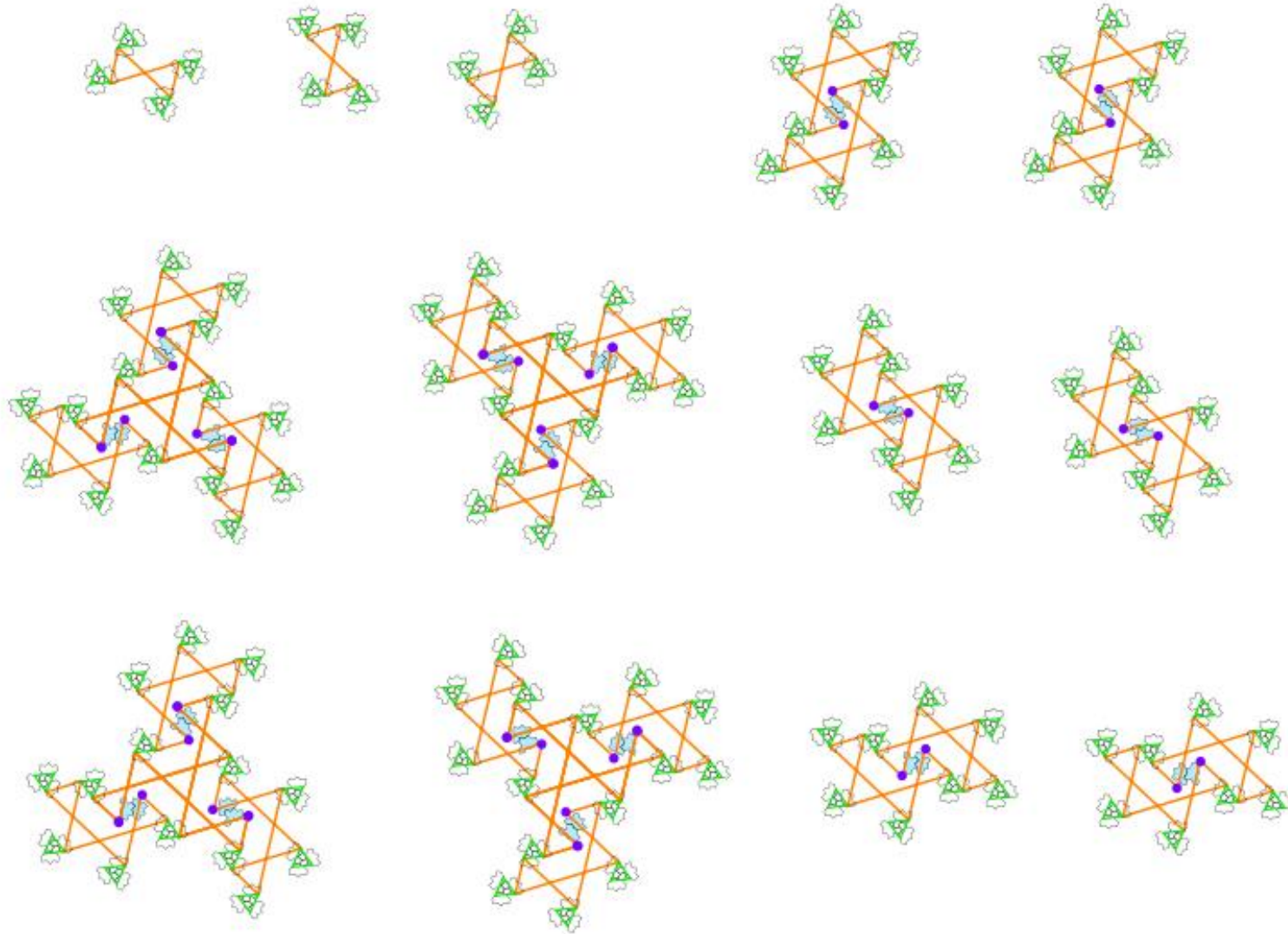
four orientations of the 3-arms



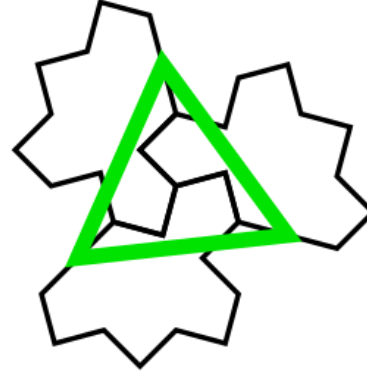
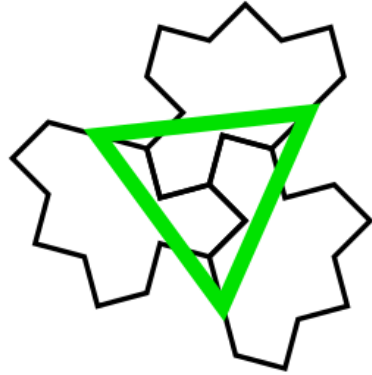
New puzzle



New puzzle

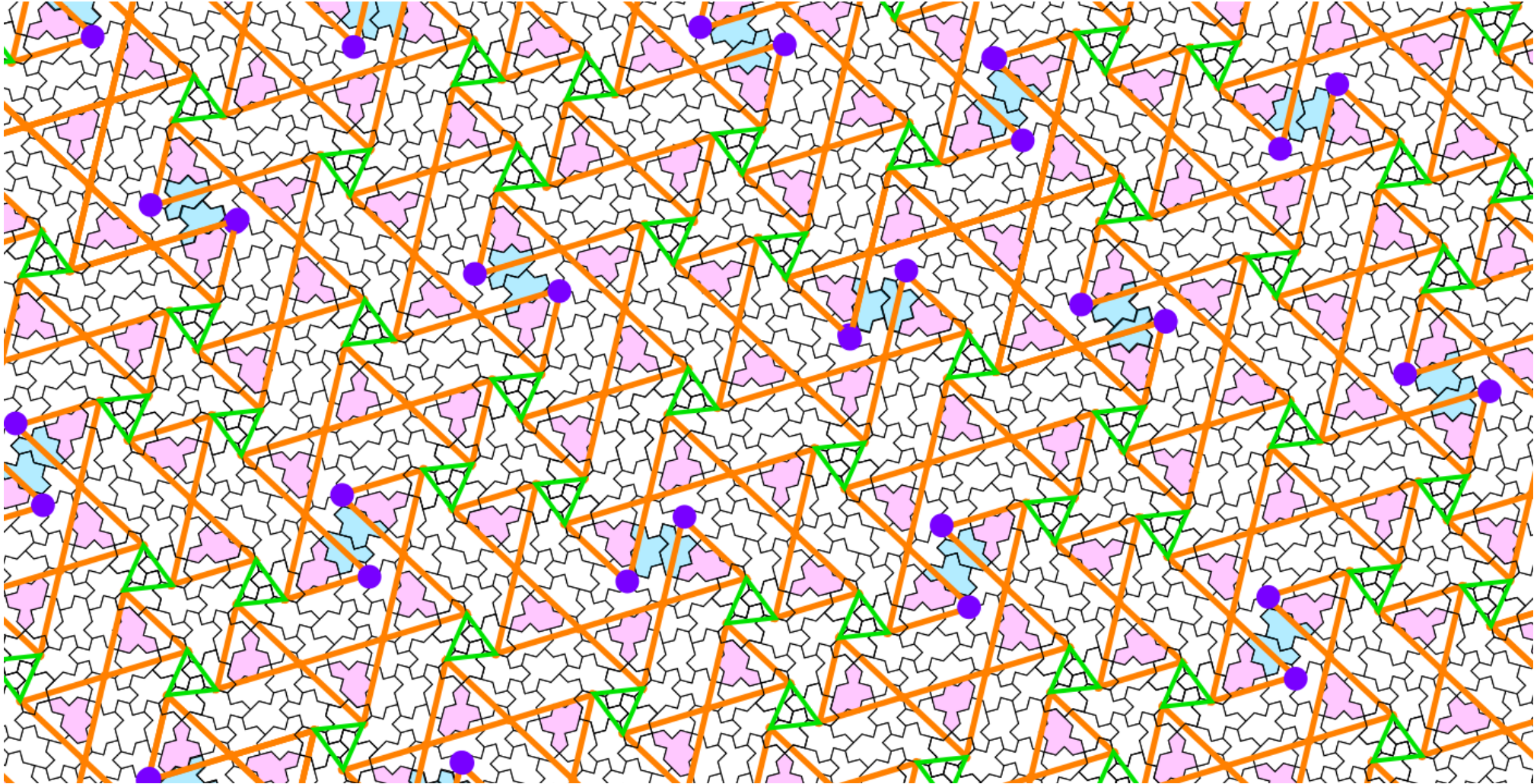


stars



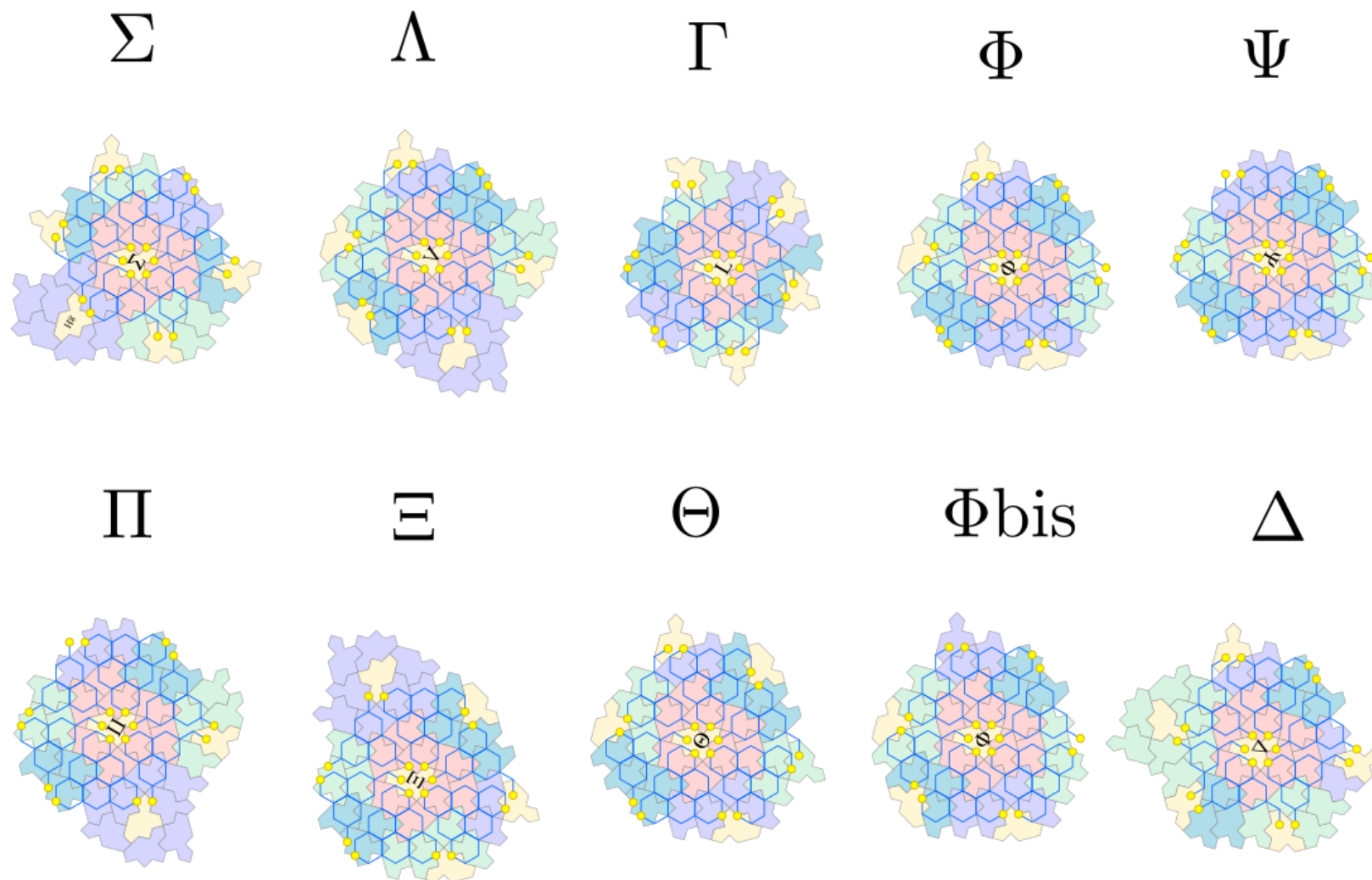
- Small and large stars with two possible orientations

Small stars connect the orange triangles



- Possibility to obtain the relative positions between diablo, 2-arms and 3-arms

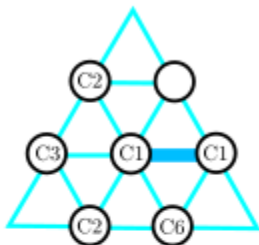
10 environments (C1 orientation)



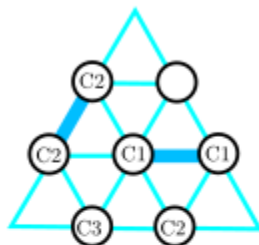
- Left handed version of the figure in Arnaud's paper.

10 environments for C1

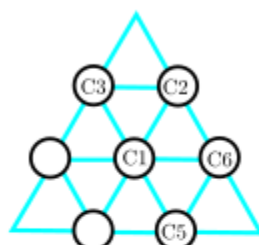
Σ



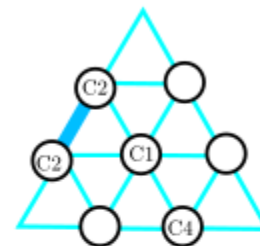
Λ



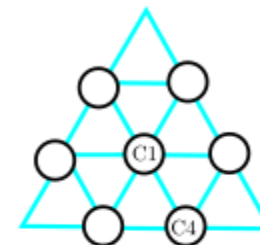
Γ



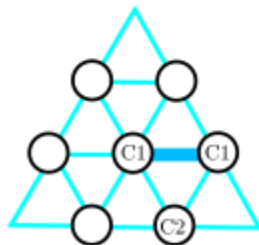
Φ



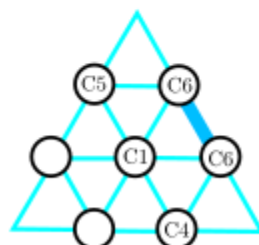
Ψ



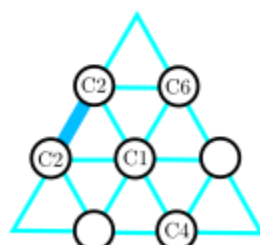
Π



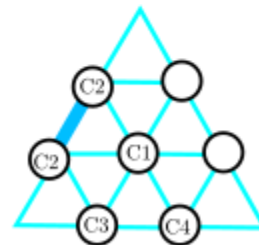
Ξ



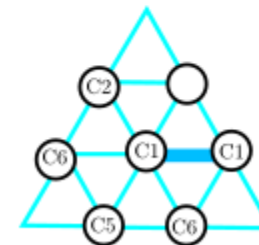
Θ



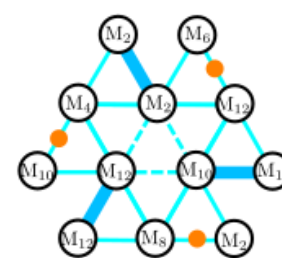
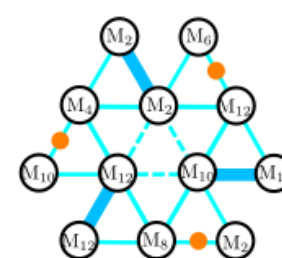
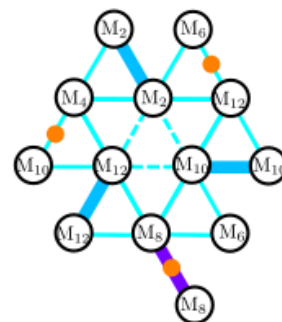
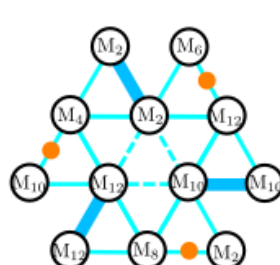
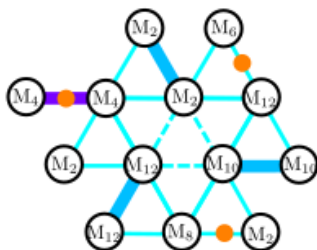
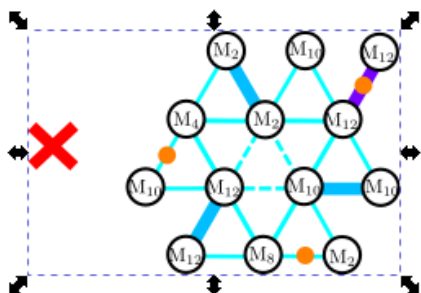
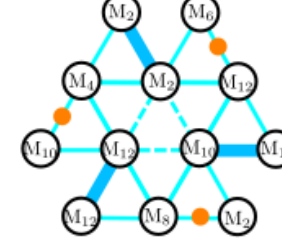
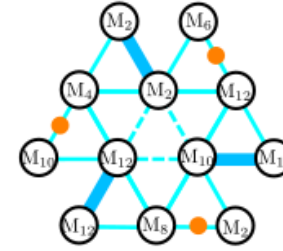
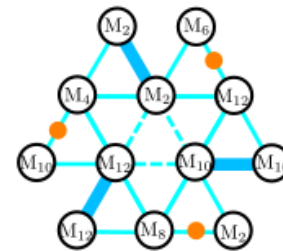
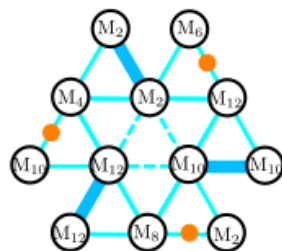
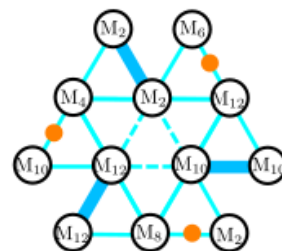
Φ bis



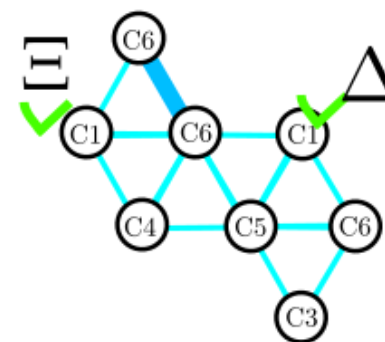
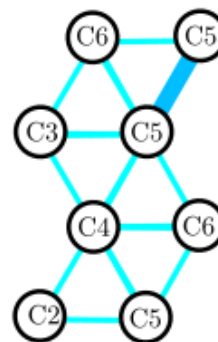
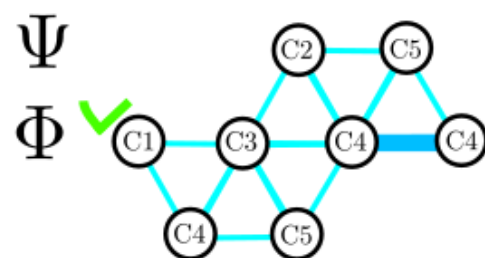
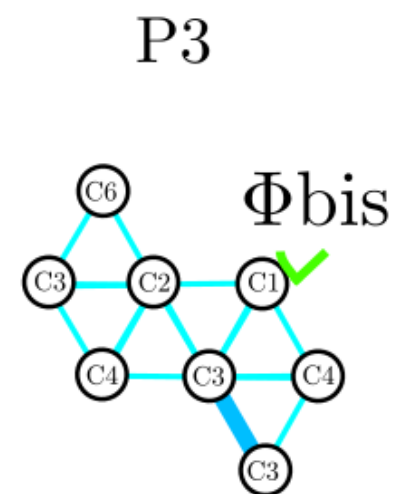
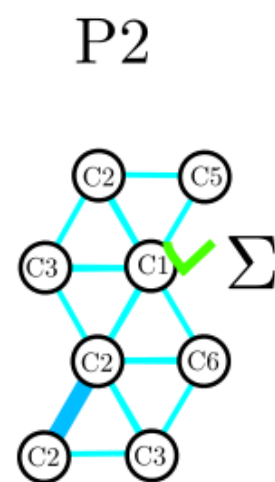
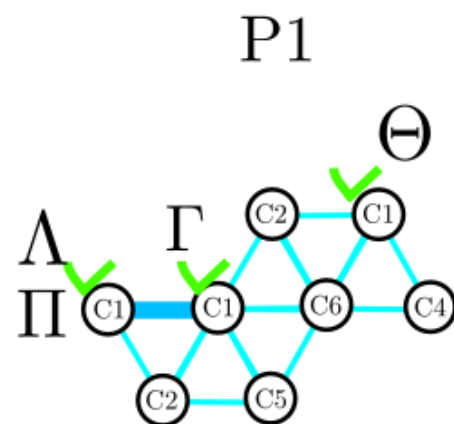
Δ



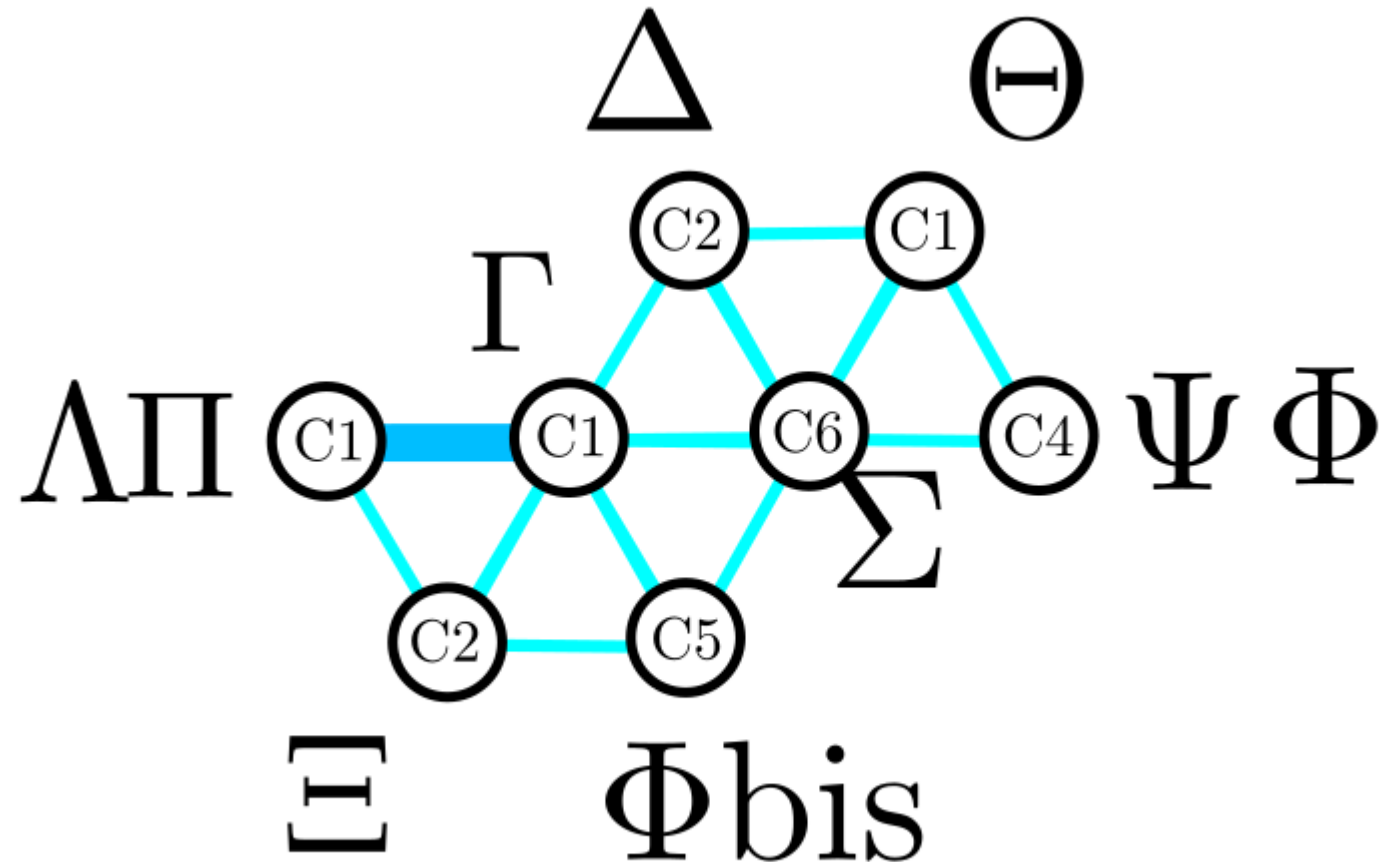
10 environments for C1

 Σ
 Λ
 Γ
 Φ
 Ψ

 Π
 Ξ
 Θ
 Φbis
 Δ


8 different locations in the chiral pattern

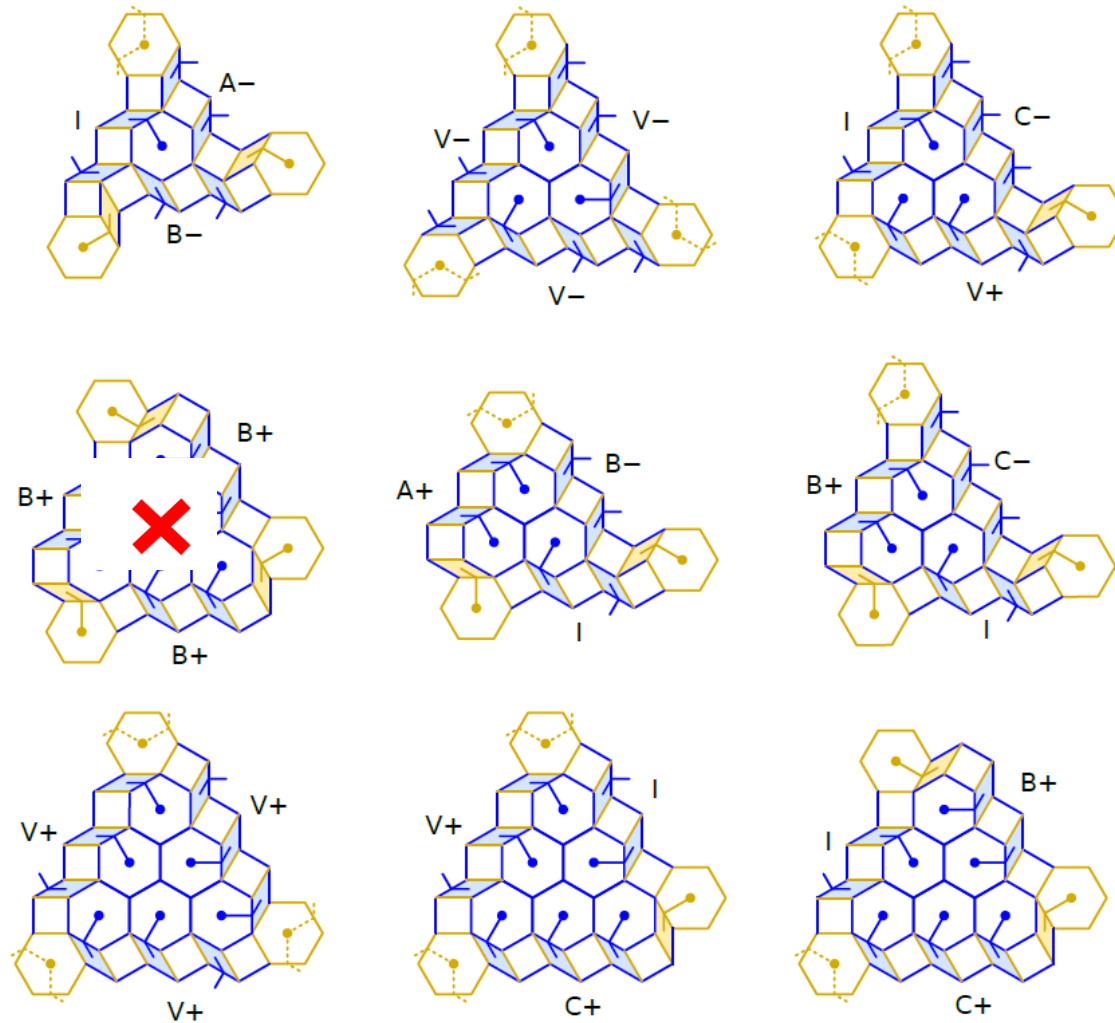


10 environments (capital greek letters)



- Need to reduce to 8 environments ?

8 types of hexagonal clusters



To do:
Locate the 8 hexagonal
clusters in the chiral pattern

Homochiral python code on github

- Under construction !
- <https://github.com/marimperorclerc/homochiral>

python code for drawing clusters with Tile(a,b) in svg graphics.

References

Code is based on the drawsvg package:

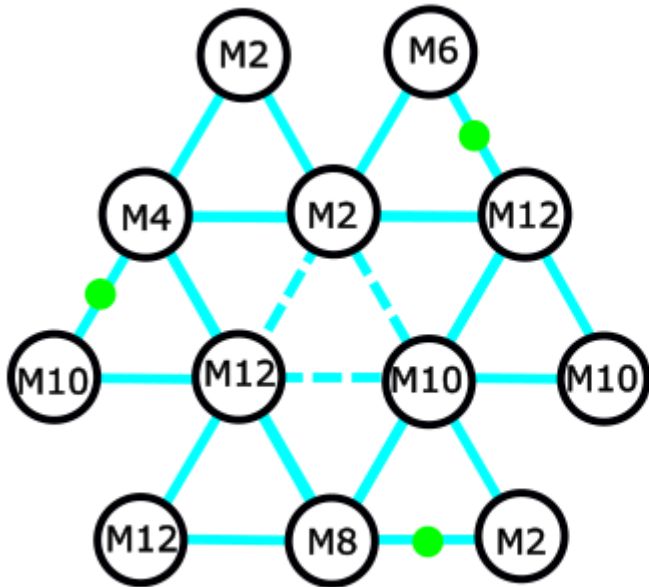
<https://pypi.org/project/drawsvg/>

And the python code with alternate chirality for inflation for spectre tiling:

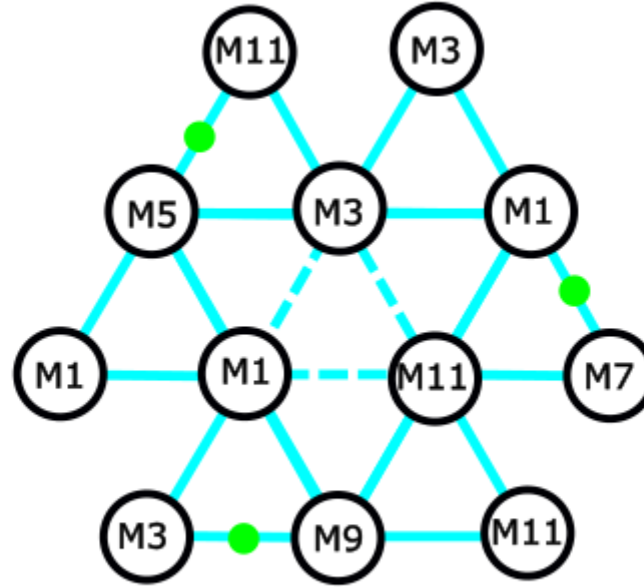
<https://github.com/shrx/spectre/blob/master/spectre.py>

Inflation step 1: substitution rule for C1

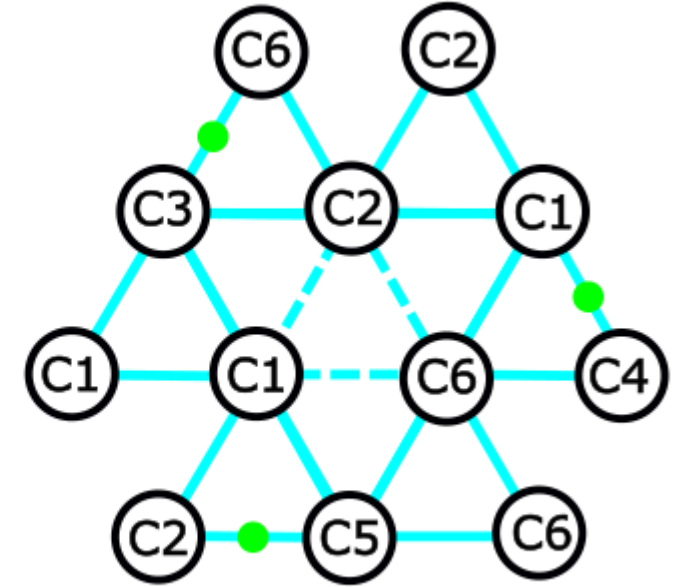
EVEN tiles in C1



ODD tiles for step 1

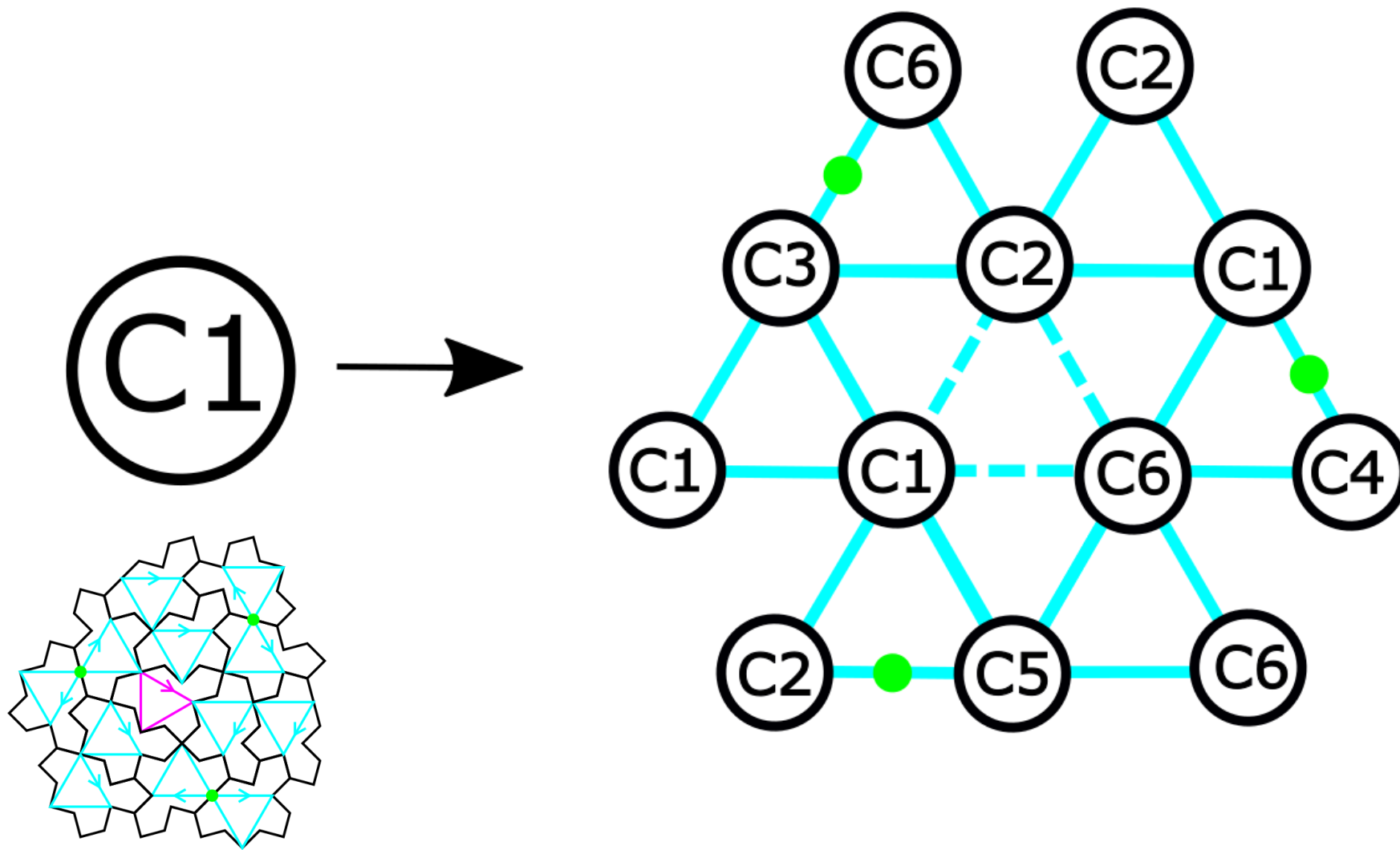


same as full clusters

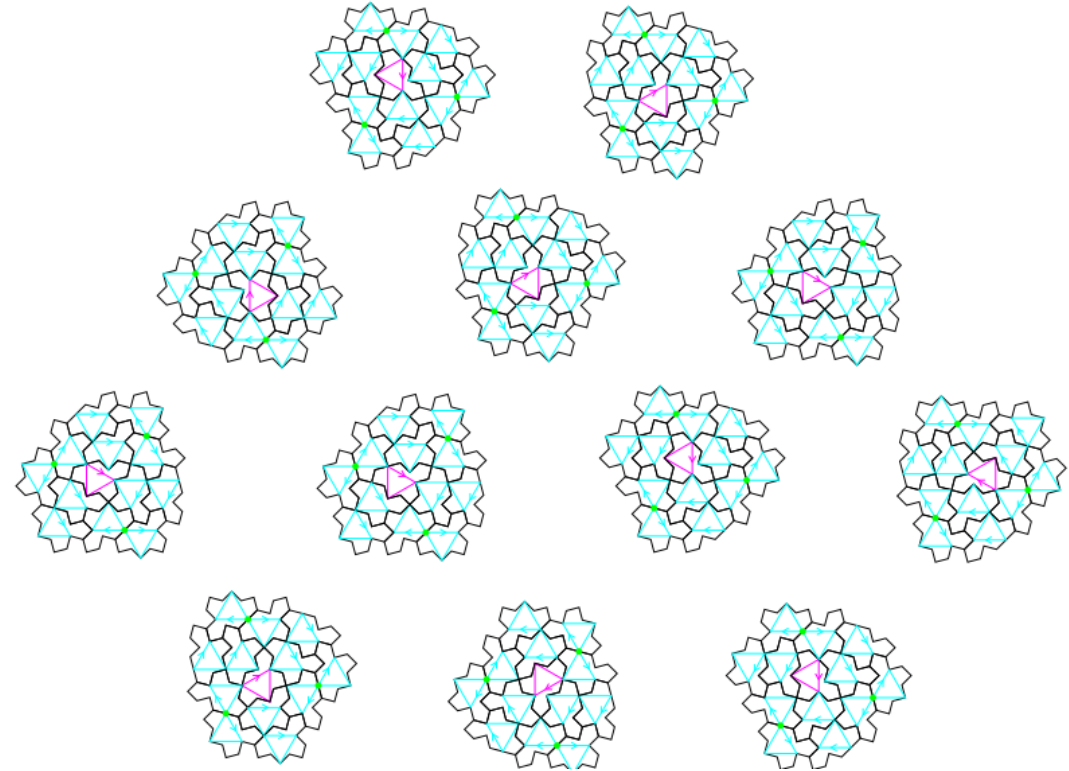
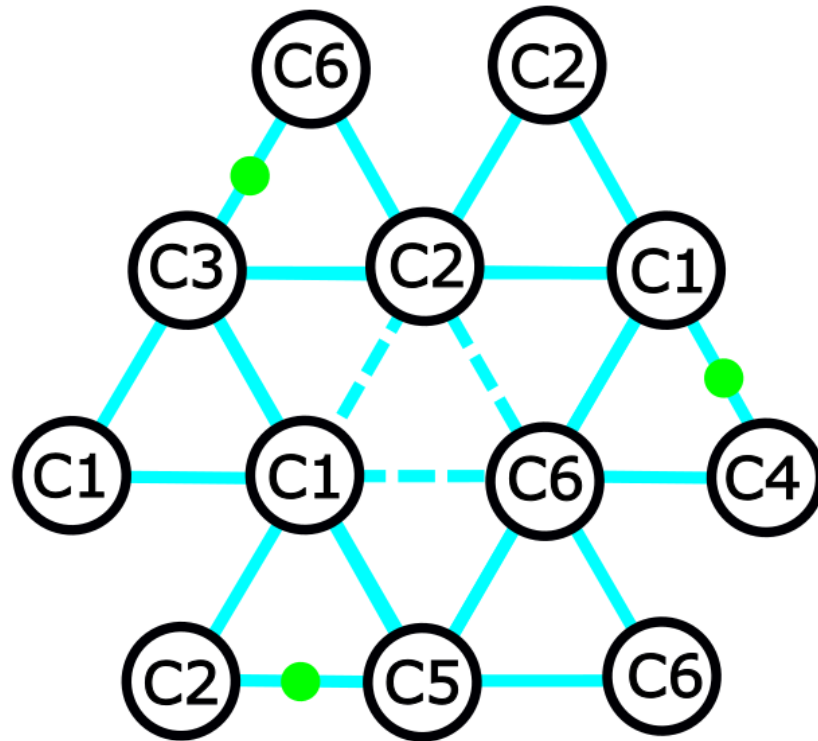


- ODD tiles for step 1 are obtained by adding 1 (mod 12) to all EVEN tiles in the C cluster (6 tiles)
- Directions of translated and half-turned tiles are exchanged (**see the change of position for green points**). Then three tiles are translated (M1, M3, M11) and three are half-turned (M5→M11, M1→M7, M9→M3)
- ODD tile index is equivalent to the orientation of a full C cluster 😊

Inflation step 1

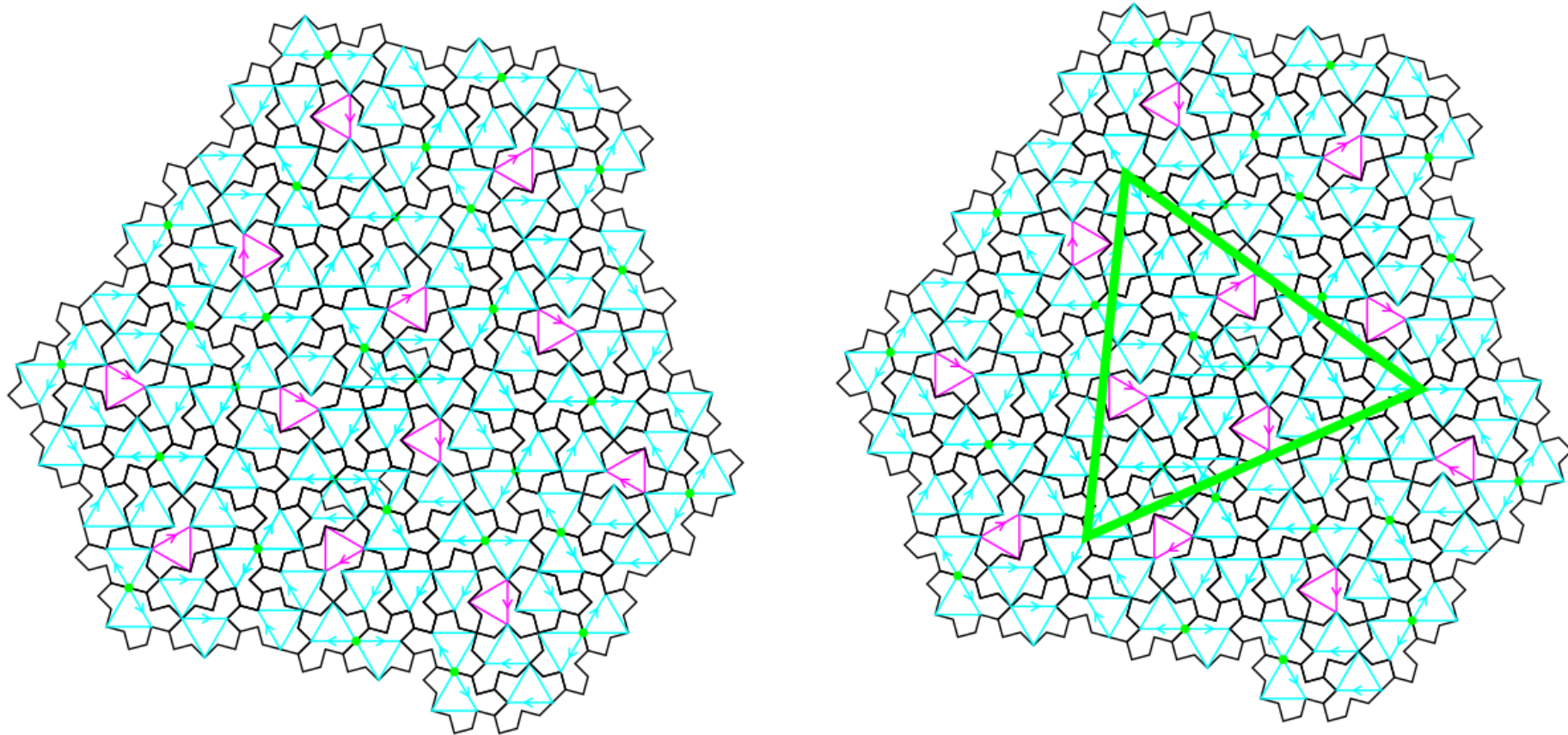


Inflation step 1



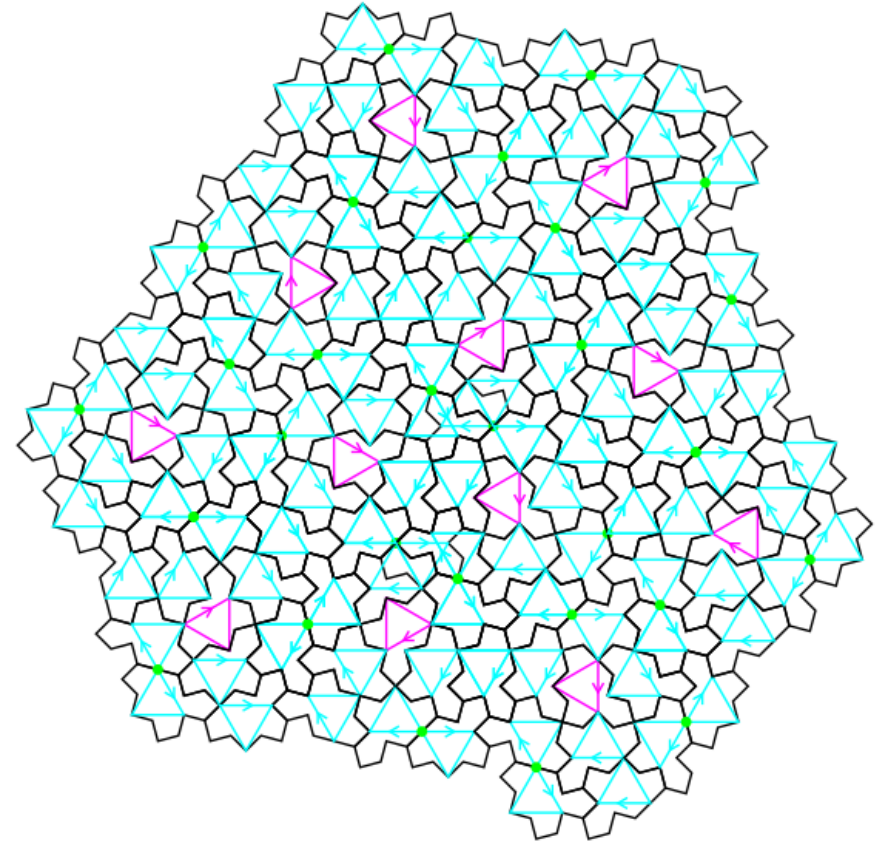
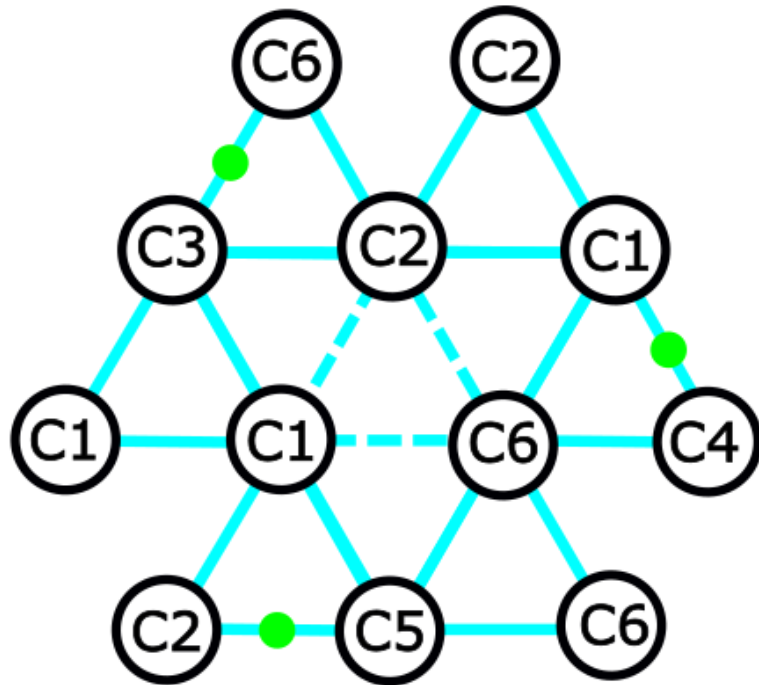
- Directions along the blue lines are the edge directions of the blue triangles !!!

Inflation step 1 for full cluster C1



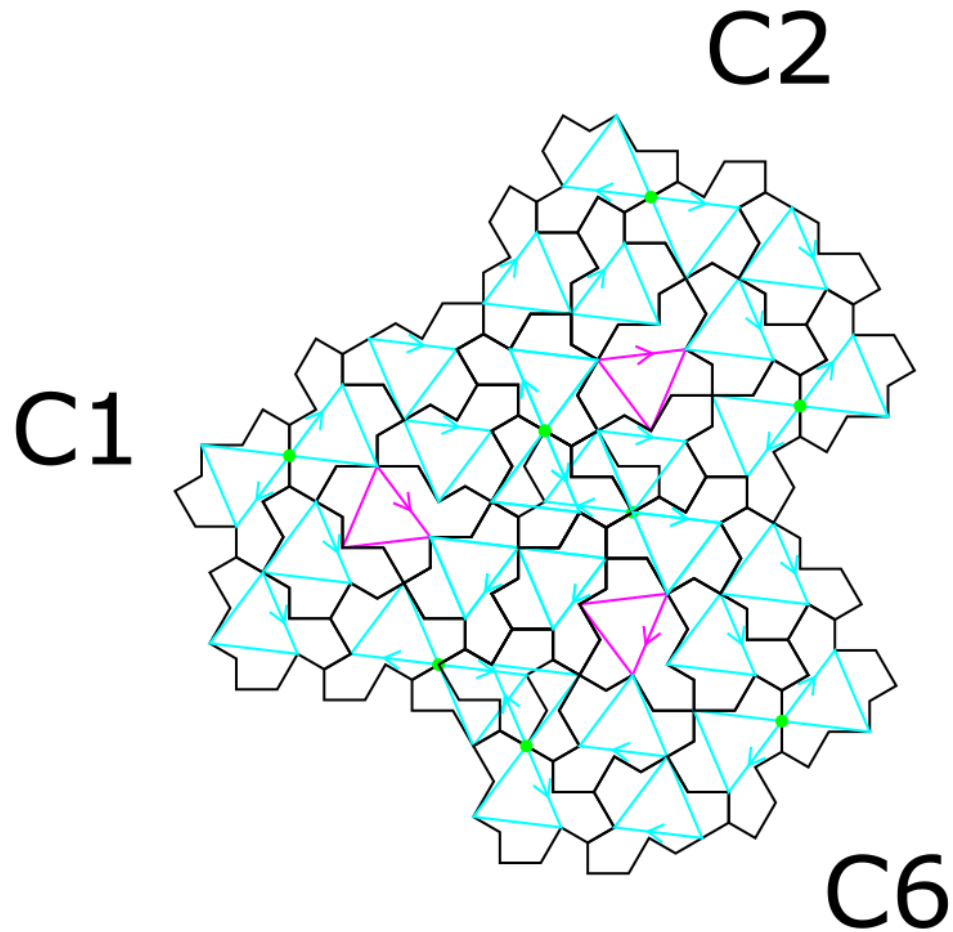
- Assembly of the 12 full clusters works fine !!! (done manually so far for the relative positions of clusters)
- Some tiles are superimposed
- Some half-turned tiles need to be removed with a simple rule: **translation wins over half-turn !**
- A bigger green triangle is identified, centered on a trifle at the center of the figure ☺

Inflation step 1 for full cluster C1

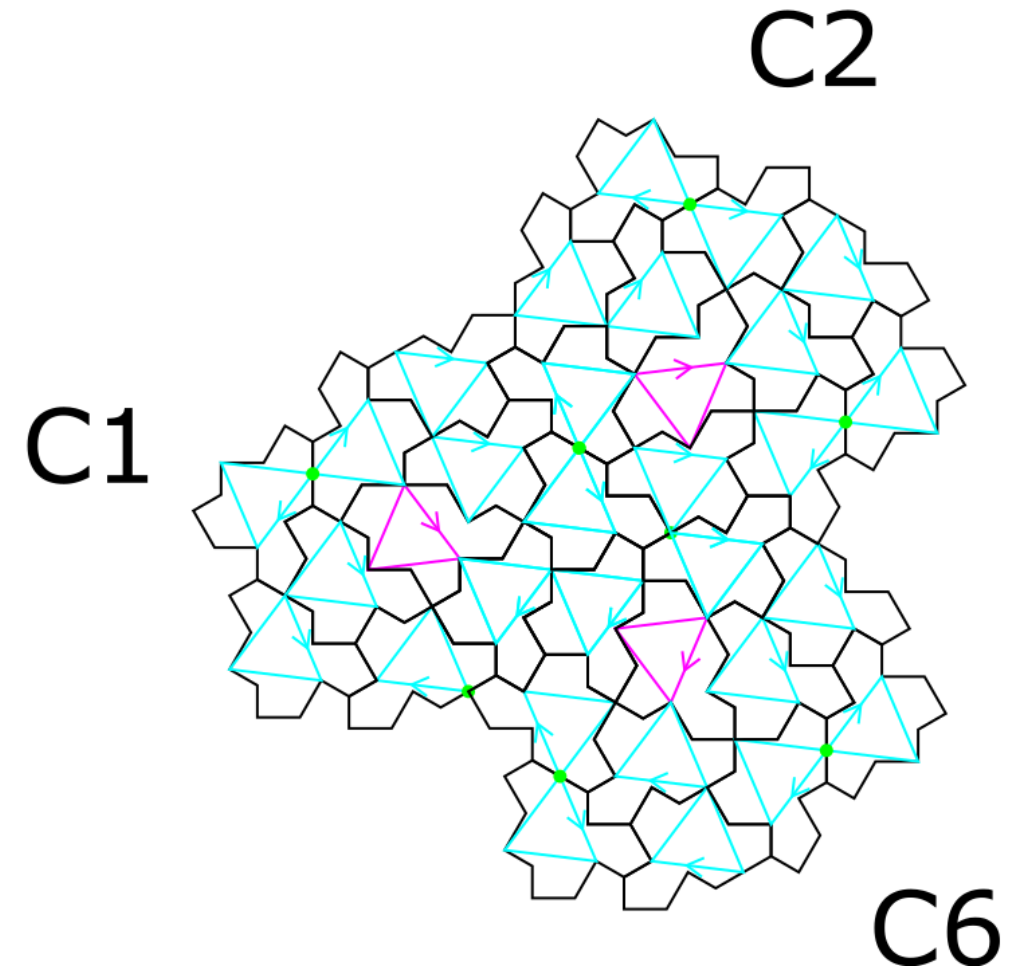


- TO DO: Find a rule for the relative positions between clusters in the assembly
- Clusters are translated from each other along the blue lines
- Use superposition of green points from two adjacent clusters ???

Defect healing

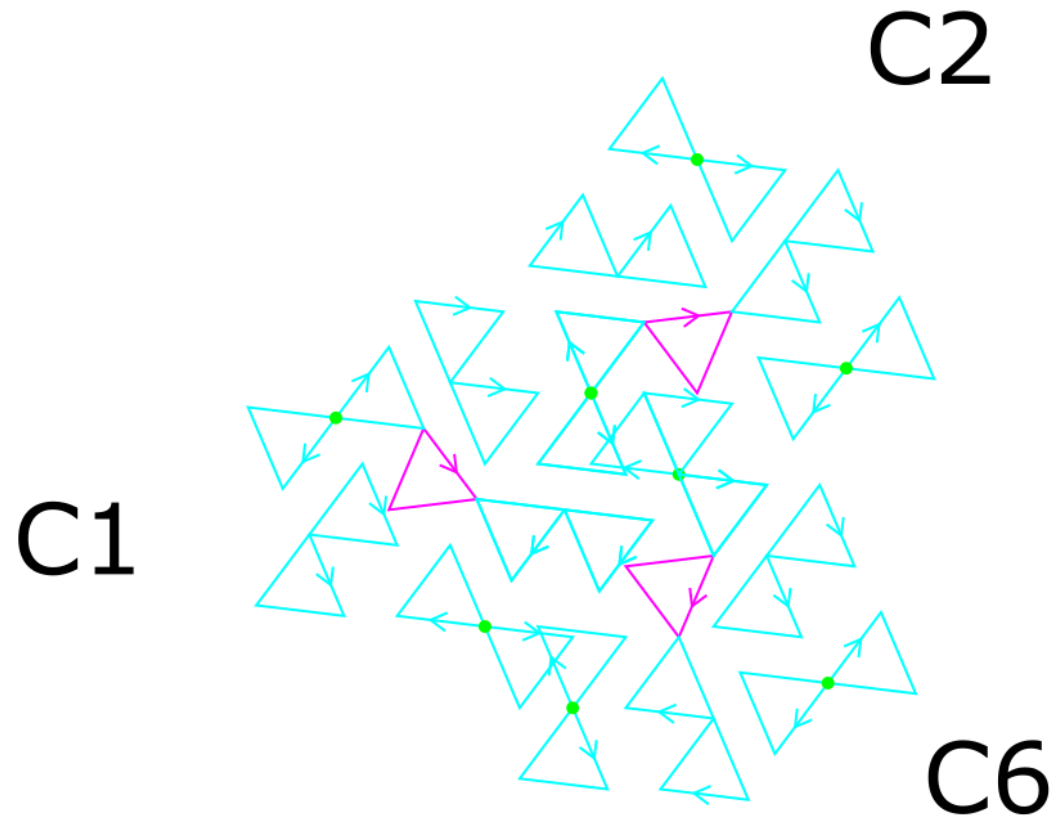


- Starting point for worms ☺

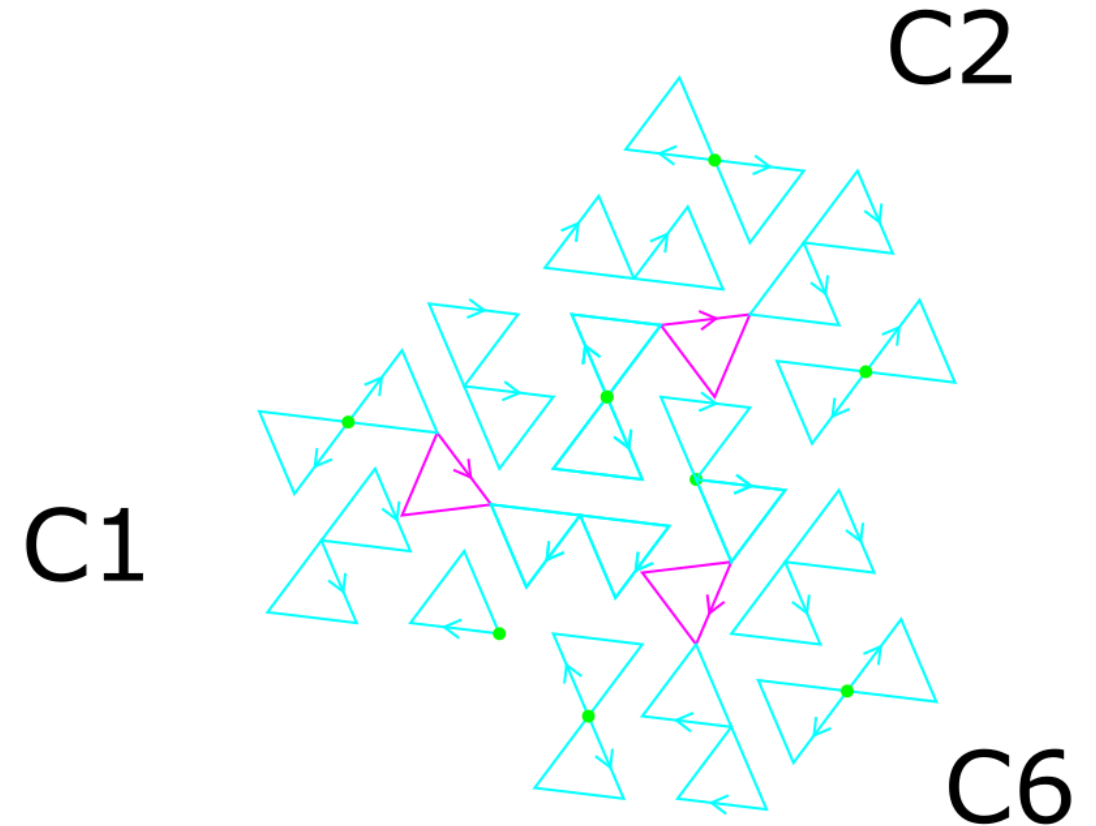


- After removing two half-turned tiles

Defect healing

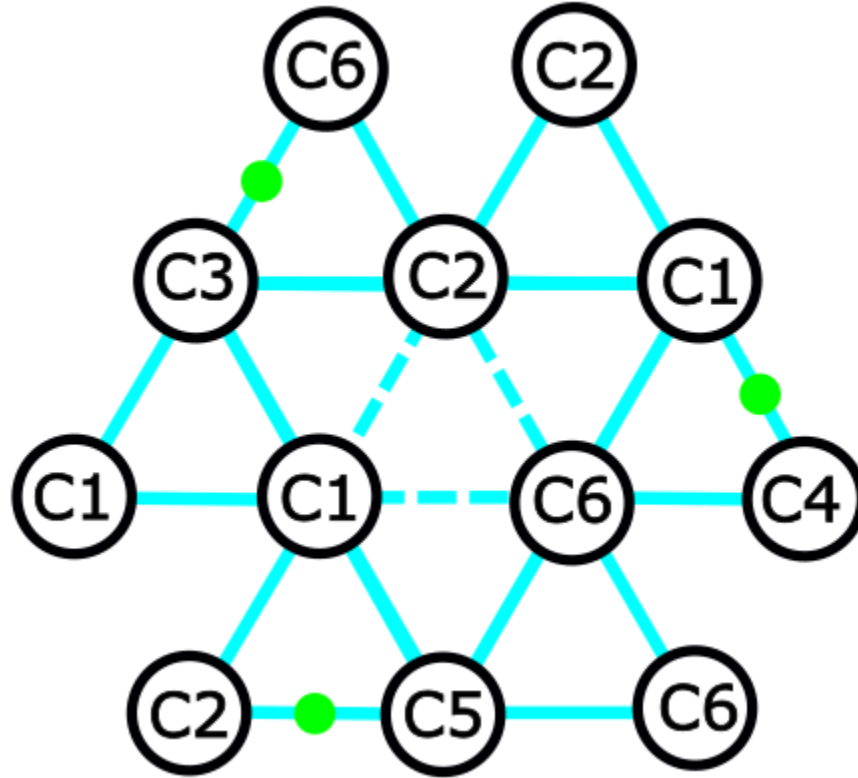


- Starting point for worms 😊



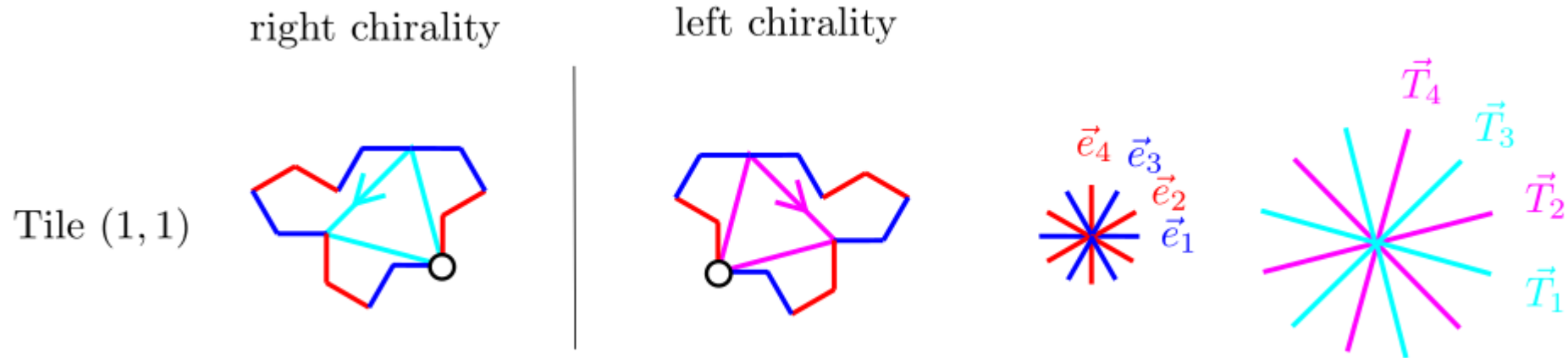
- After removing two half-turned tiles

TO DO: Step 2



- Replace all the 12 full clusters in this pattern with the same rule used previously for C1 (just a rotation of the pattern for C2, C3, C4, C5 and C6)

New reference triangle in Tile(a,b)

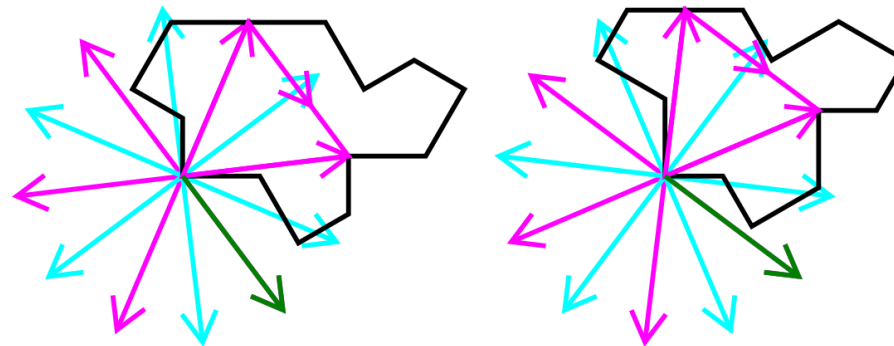


Robust for any T(a,b) tile ☺

Tile(1.6,1.2)

Tile(1.2,1.6)

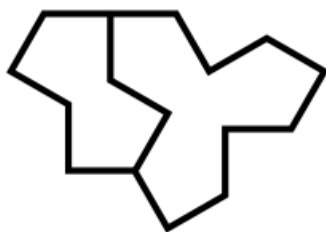
$a=1.6$
 $b=1.2$



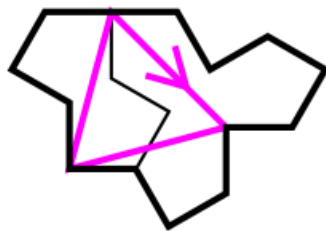
Link with trifle and bow-tie in $\text{Tile}(1,1)$

Tile (1, 1)

bow-tie



trifle



half bow-tie



half trifle

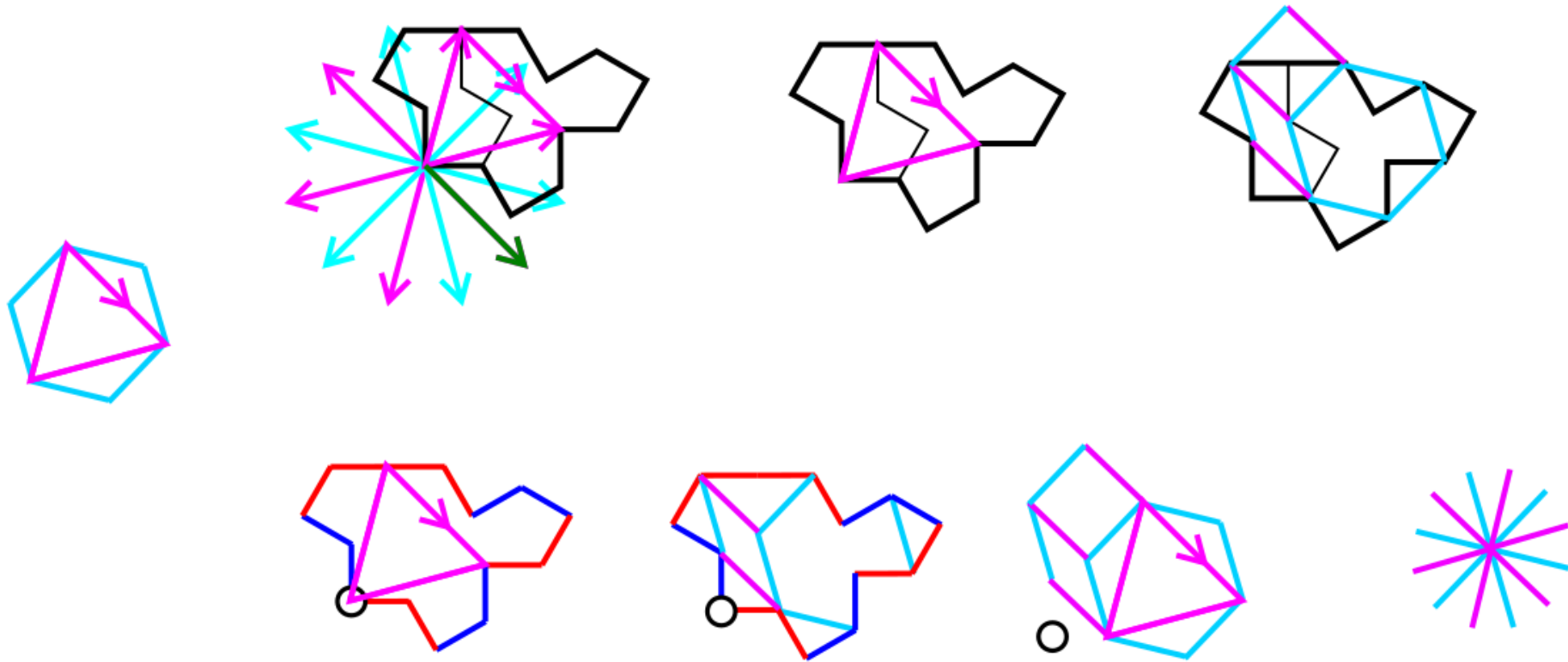


half bow-tie



Comparison with hexagon/square/rhombus decomposition

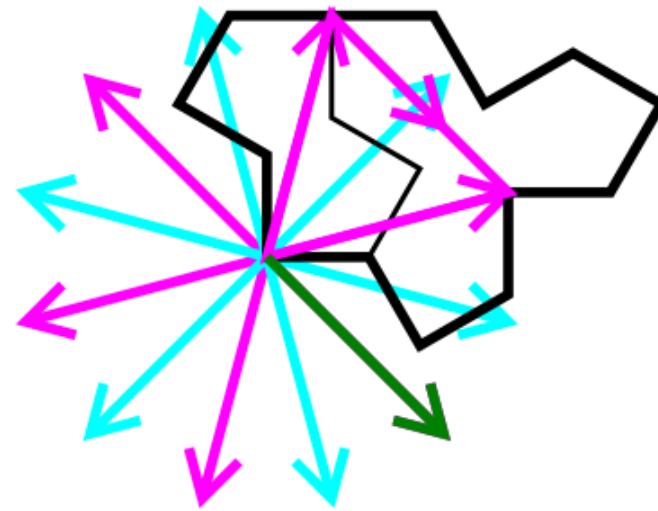
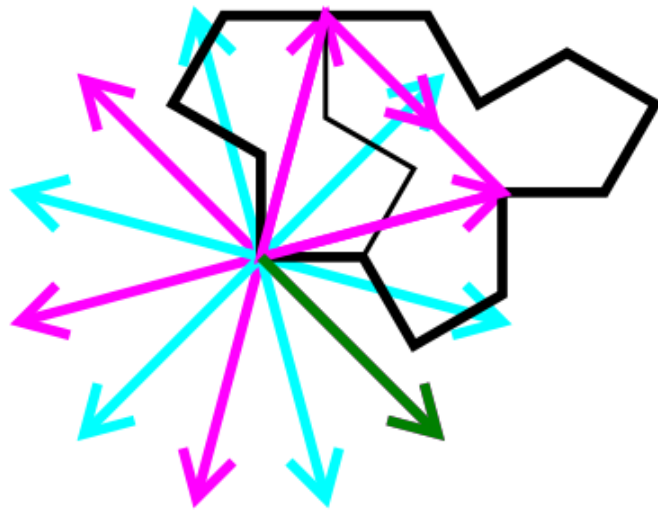
Tile(1.0,1.0)



Tile(1,1) : spectre tile

Tile(1.0,1.0)

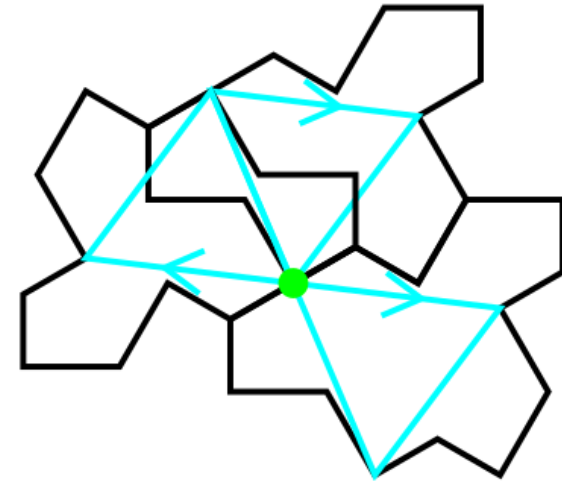
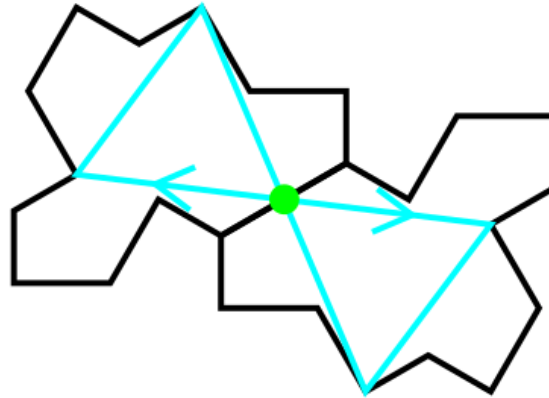
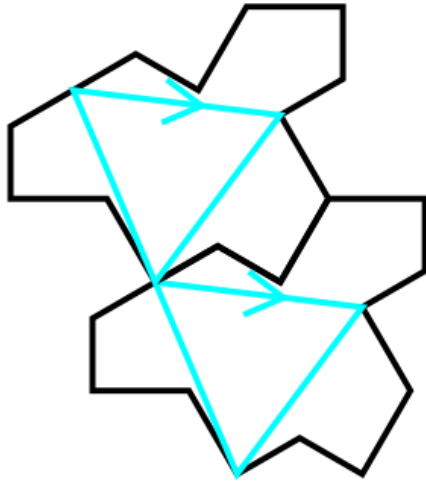
Tile(1.0,1.0)



translation versus half-turn for any $\text{Tile}(a,b)$

$a=1.6$

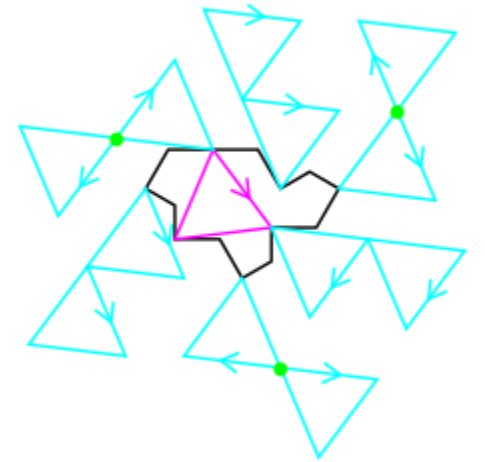
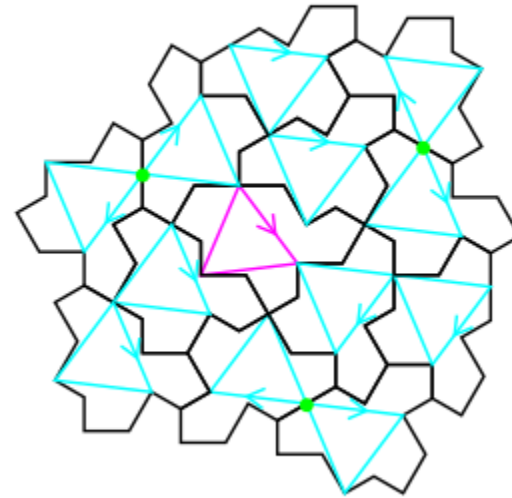
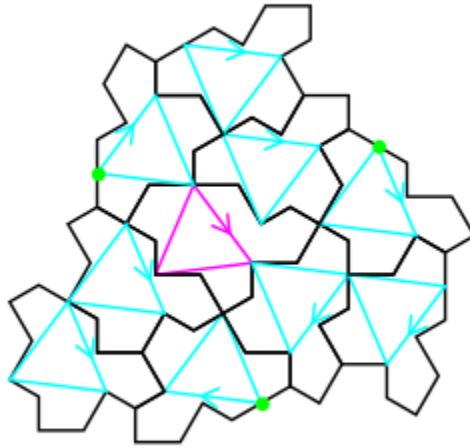
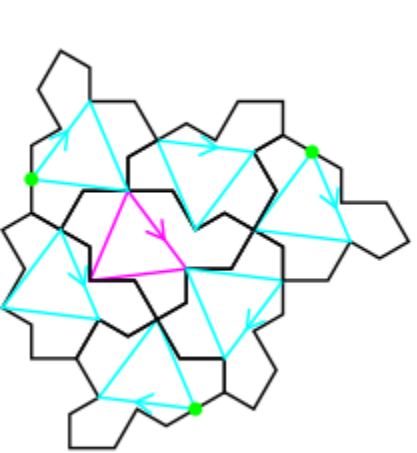
$b=1.2$



Cluster and full cluster

$$a=1.6$$

$$b=1.2$$



C cluster
(corona around a
ODD tile)

The three
translated tiles
are all in the tiling

Three more possible tiles
with half-turn around the
green points